LKA-GFNet: Language Knowledge-Augmented Graph Fusion for Tri-Source Heterogeneous Remote Sensing Data Classification

Wei Cheng, Ruyi Li, Junheng Zhu, and Xianghai Wang

***Abstract*—Remote sensing (RS) image classification is pivotal for fine-grained land-cover mapping and environmental monitoring. Recently, multisource data fusion has emerged as a mainstream approach for enhancing classification accuracy. However, current fusion-based classification methods remain largely limited to dual-source integration, leaving the richer complementary information inherent in tri-source data—such as hyperspectral (HS), multispectral (MS), and radar data—underexplored. Furthermore, existing methods rely almost exclusively on image features and lack systematic incorporation of the textual semantic priors derived from geoscience classification standards. This oversight hinders the model's capacity to capture the deeper associations between image features and land-cover concepts, thereby considerably limiting classification performance. To address these gaps, this article proposes a language knowledge-augmented graph fusion network (LKA-GFNet) for the fusion classification of HS, MS, and radar data. By incorporating textual semantic priors of geoscience labels into tri-source feature learning and alignment, LKA-GFNet enables semantics-driven high-accuracy classification. Specifically, a strong-weak dual-stage alignment framework is introduced to enable progressive image semantic perception, leveraging unified textual priors as shared references to collaboratively optimize both geoscientific knowledge and tri-source heterogeneous image features. Concurrently, we design a compound threshold-constrained graph-driven decision fusion (CT-GDF) module for interactive fusion of tri-source heterogeneous features. It constructs intramodal and cross-modal graphs via cumulative probability and adaptive distance thresholds, mining latent complex dependencies and uncertainties within the tri-source data. Extensive experiments on four tri-source datasets demonstrate that LKA-GFNet significantly outperforms state-of-the-art (SOTA) methods, achieving robust land-cover classification with enhanced interpretability. The source code will be uploaded to https://github.com/cwlnnu/LKA-GFNet.**

***Index Terms*—Fusion classification, graph fusion, language knowledge, semantic priors, tri-source remote sensing (RS) data.**

Received 3 February 2026; accepted 25 February 2026. Date of publication 27 February 2026; date of current version 24 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 42371338 and in part by the Key Scientific Research Project of Liaoning Provincial Education Department under Grant JYTZD2023101. *(Corresponding author: Xianghai Wang.)*

Wei Cheng and Junheng Zhu are with the School of Geographical Sciences, Liaoning Normal University, Dalian 116029, China (e-mail: chengsolstice@163.com; junheng1020@163.com).

Ruyi Li is with the School of Computer Science and Artificial Intelligence, Liaoning Normal University, Dalian 116029, China (e-mail: 2314555010@qq.com).

Xianghai Wang is with the School of Geographical Sciences and the School of Computer Science and Artificial Intelligence, Liaoning Normal University, Dalian 116029, China (e-mail: xhwang@lnnu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3668976

NOMENCLATURE

$X_{HS}, X_{MS}, X_{Radar}$	Tri-source images.
$B; H, W$	Batch size; height and width of images.
$L; C, D$	Sequence length; features dimension.
T, F	Text and image features.
$D(i,j), P(i,j)$	Distance and probability matrices.
$\mathscr{T}(i,j)$	Dynamic distance threshold.
$A(i,j)$	Adjacency matrices.
θ, α, β	Hyperparameters.
R_1, R_2, R_3	Classification results.
$Y; \mathscr{L}$	Label; loss functions.

I. INTRODUCTION

REMOTE sensing (RS) image classification is fundamental to intelligent RS interpretation and underpins many Earth science applications [1]. With the rapid growth of RS data, the field is shifting from unimodal to multimodal classification to better characterize complex land-cover types, benefiting land-cover mapping, land-use surveying, ecological monitoring, and disaster assessment [2], [3], [4].

Multisource RS fusion is motivated by the complementary strengths of different sensors: LiDAR/SAR provide reliable structural cues but limited spectral discrimination [5], hyperspectral (HS) data offer rich spectra for material identification but suffer from coarse spatial resolution [6], and multispectral (MS) data retain spatial details yet struggle with spectrally similar classes [7]. Consequently, single- or even dual-source methods may fail in complex scenes, while tri-source integration (HS-MS-SAR/LiDAR) enables more robust, multiscale surface characterization.

Early multisource fusion methods largely followed conventional handcrafted paradigms. Rasti et al. [8] used sparse low-rank modeling to derive fused low-rank representations from HS and LiDAR, which were then classified by a downstream classifier. Hong et al. [9] aligned common components across HS-LiDAR-SAR to decompose them into shared and modality-specific subspaces, and followed by nearest-neighbor classification. Hänsch and Hellwich [10] introduced a variant random forest that can be directly applied for joint classification of HS, RGB, and LiDAR data. These methods often rely on handcrafted features, making it difficult to fully capture complex cross-modal relationships and rendering them susceptible to noise.

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

In recent years, deep learning (DL) has become the dominant paradigm for multisource RS classification due to its powerful nonlinear modeling capacity. Wang et al. [11] leveraged the spectral richness of HS data and the spatial complementarity of LiDAR to achieve synergistic optimization of spectral and spatial features. Xu et al. [12] developed a fully convolutional network for fusing HS, LiDAR, and very-high-resolution imagery. Pan et al. [13] employed attention mechanisms to recalibrate features across distinct categories and proposed an attribute-guided fusion strategy to explore intermodal correlations. Xue et al. [14] adopted an encoder–decoder architecture to extract key information from multimodal data and utilized cross-attention to enhance intermodal complementarity.

With ongoing advances in DL, graph convolutional networks (GCNs) and transformer architectures have also emerged as key research directions due to their capability for global context modeling. Wang et al. [15] constructed multiscale spatial graphs across modalities and captured shared class-discriminative information via dynamic graph aggregation. Yang et al. [16] developed a multifrequency GCN that leverages high- and low-frequency cues from HS and LiDAR to enhance multimodal features. Zhang et al. [17] reformulated multiview subspace clustering for HS-LiDAR clustering as a link-prediction problem on a sparse multimodal attribute graph. Du et al. [18] introduced a spectral–spatial graph network that jointly models local and global HS-LiDAR correlations through a convolutional branch and a spectral–spatial graph branch. Jing et al. [19] learned dynamic heterogeneous graph structures to model cross-modal topological relations for efficient interactive fusion, leading to more robust classification performance. Zhao et al. [20] proposed a fractional Fourier image transformer to extract global contextual information from multisource data, achieving higher efficiency than standard transformers. Hu et al. [21] developed a cross-memory quaternion transformer to model both intramodality and intermodality correlations, thereby learning more discriminative fused representations. Huang et al. [22] designed a hybrid cross-patch attention mechanism in a multimodal fusion transformer, where an external CLS token generated from LiDAR is jointly used with HS patch tokens to strengthen feature representations. Zhu et al. [23] proposed a multihead axial-attention transformer that effectively fuses the spatial–spectral features of HS with LiDAR elevation cues, and captures multiscale information via axial attention. Recently, Mamba, an emerging sequence modeling architecture, has attracted significant attention. Gao et al. [24] proposed a multiscale Mamba feature fusion network that enhances features through spatial and spectral Mamba modules, while incorporating a fusion Mamba module to explore intermodal feature interactions. Ma et al. [25] introduced a multipath scanning Mamba to capture global features across spatial and spectral dimensions, achieving cross-modal feature alignment via a channel interaction strategy.

In addition, several dual-source methods can be extended to tri-source scenarios. For example, Lu et al. [26] proposed a coupled adversarial learning method with two parallel branches to capture detailed multimodal cues, which can be naturally generalized to a three-branch design. Wang et al. [27] used a symmetric dual-branch architecture and enhanced multimodal representations with graph structures derived from radar data, which is also applicable to tri-source feature modeling. Xie et al. [28] fused multisource features via element-wise addition, a strategy that can be directly extended to tri-source data. Despite their extendability, these methods do not systematically explore the complementarity of heterogeneous tri-source data. Moreover, although existing tri-source fusion classification studies (see [9], [12]) consider three inputs, their core fusion often still relies on simple element-wise addition or concatenation, without explicitly modeling cross-modal correspondences and semantic consistency. In contrast, guided by geoscience label semantics, our method systematically explores correlations among heterogeneous tri-source data and constructs robust graph connections for deep intersource interaction and fusion, thereby fully leveraging their complementary strengths. Meanwhile, several studies have directly fused three sources. Iervolino et al. [29] combined panchromatic, MS, and SAR images using intensity-hue-saturation transformation and multiresolution analysis. Hong et al. [30] fused HS, MS, and SAR for cross-scene semantic segmentation with an emphasis on domain generalization. Li et al. [31] integrated MS, HS, and LiDAR for mangrove mapping, and Meng et al. [32] combined panchromatic, MS, and HS for super-resolution. These tri-source works are typically task-specific, and their objectives and fusion mechanisms differ substantially from our classification task.

Therefore, multisource fusion classification still suffers from the following limitations.

1) *Lack of Systematic Tri-Source Synergistic Modeling:* While dual-source fusion (e.g., HS-LiDAR) has shown strong performance, it captures only partial complementary cues, such as spectral-elevation coupling. Dual-source fusion can still struggle when classes share similar spectra and elevations, leading to limited discriminability and semantic drift. In contrast, HS, MS, and radar provide complementary spectral, spatial-textural, and structural-scattering information, and their joint modeling can yield a more holistic spectral–spatial-structural representation. Therefore, there is an urgent need to develop fusion frameworks capable of systematically modeling the intrinsic relationships among tri-source data to improve recognition and classification in complex heterogeneous scenes.
2) *Underutilization of Geoscience Knowledge-Driven Textual Priors:* Existing studies rely almost exclusively on spectral or spatial features inherent in images and rarely integrate label-text semantics as a complementary prior in a systematic manner. As a result, the learning process lacks high-level semantic guidance, making it difficult to capture deeper associations between image features and the intrinsic properties of land-cover classes. The class names in benchmark classification datasets are not arbitrary symbols. Instead, they constitute concept sets defined by geoscience experts based on knowledge of land-cover types, ecological processes, and other domain principles. As such, the labels themselves encode

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

implicit geoscientific priors. Consequently, integrating geoscience textual priors to move beyond image-only learning and jointly optimize multimodal alignment remains a key challenge.

In response, the core contribution of this study is to overcome the limitations of existing dual-source fusion methods and construct a unified cross-modal learning framework that systematically integrates geoscience prior knowledge into the tri-source fusion. Specifically, this article proposes a language knowledge-augmented graph fusion network (LKA-GFNet) for the fusion classification of HS, MS, and radar data. The proposed progressive semantic-aware alignment (PSAA) module establishes a novel strong-weak dual-stage alignment paradigm. It utilizes the semantic space constructed from textual priors as supervision to drive structured semantic reconstruction of image feature spaces, progressively aligning heterogeneous tri-source features into a unified semantic representation. Furthermore, a compound threshold-constrained graph-driven decision fusion (CT-GDF) module is proposed to facilitate in-depth interactions among tri-source heterogeneous features. This module constructs intramodal and cross-modal graphs using a compound thresholding mechanism based on cumulative probability and dynamic distance, enabling efficient information propagation within and between modalities. Ultimately, classification results are obtained via decision fusion. The main contributions of this work are as follows.

1) An LKA-GFNet is proposed for the fusion classification of HS, MS, and radar data. Under the guidance of geoscience textual prior knowledge, LKA-GFNet explores deeper synergistic and complementary relationships among tri-source heterogeneous features, achieving knowledge-driven high-precision classification within a unified cross-modal learning framework.
2) A PSAA module is designed to establish a strong-weak dual-stage alignment paradigm. This module utilizes textual prior knowledge as unified high-dimensional semantic anchors, providing a shared semantic reference framework for tri-source heterogeneous features. By aligning image features with consistent semantic prototypes, it mitigates multimodal distribution shifts and achieves knowledge-driven cross-modal semantic consistency.
3) A CT-GDF module is developed to construct dynamic intramodal and cross-modal graphs. Using a compound thresholding mechanism based on cumulative probability and dynamic distance, it adaptively screens robust feature associations while eliminating noisy connections and redundant edges. Discriminative tri-source information fusion and reliable decision outcomes are achieved through the integration of intramodal and cross-modal graphs.
4) Systematic experiments on four public tri-source RS datasets validate the effectiveness of LKA-GFNet. The framework outperforms state-of-the-art (SOTA) algorithms across multiple evaluation metrics, demonstrating robust performance across diverse land-cover types while exhibiting enhanced interpretability. Extensive ablation studies further confirm the efficacy of LKA-GFNet and reveal substantial performance advantages.

The remainder of this article is structured as follows. Section II introduces image–text multimodal learning and graph construction. Section III elaborates on the architectural details of LKA-GFNet. Section IV presents experimental results and analysis. Section V provides concluding remarks.

II. Related Work

A. Image–Text Multimodal Learning

Image–text multimodal learning seeks to integrate visual and linguistic information into a unified representation space, thereby enhancing cross-modal understanding and generation. A core challenge in this field, compared with unimodal learning, lies in mitigating cross-modal heterogeneity, semantic granularity mismatch, and expression divergence. The contrastive language-image pretraining (CLIP) model has established a foundational paradigm for image–text multimodal learning through contrastive learning, aligning images and texts in a shared semantic space and demonstrating robust zero-shot recognition and cross-modal retrieval capabilities [33].

In the field of RS, image–text contrastive learning provides a means to leverage rich linguistic priors in addition to RS data, facilitating semantic alignment between geographic concepts and visual features. Liu et al. [34] proposed RemoteCLIP, a vision-language foundation model for RS that learns semantically rich visual features aligned with textual descriptions, enabling effective transfer to downstream tasks. Li et al. [35] introduced a contrastive vision-language supervised model for RS scene classification. Pretrained on a large-scale image–text dataset, the model employs contrastive loss to learn semantically rich visual representations. Zhang et al. [36] developed a language-aware domain generalization network that uses coarse- and fine-grained linguistic features to construct a shared semantic space across domains. Through contrastive alignment of visual and linguistic representations, the model achieves reliable cross-scene classification. Dong et al. [37] proposed ChangeCLIP for RS change detection, adapting CLIP to extract bi-temporal representations. It further injects image–text embeddings into decoder features to enhance semantic representation.

Although textual priors have proven effective in enhancing visual representations in RS, the systematic modeling and integration of such knowledge—particularly from geoscience—remain underexplored in multisource fusion classification. Most existing efforts are limited to single- or dual-source scenarios, leaving the potential of language-aware joint modeling in tri-source heterogeneous data largely untapped. Therefore, effectively integrating linguistic priors within a tri-source fusion framework to enable collaborative image–text and multimodal feature learning remains an open research challenge.

B. Graph Construction

Graph construction is a fundamental step in graph representation learning, especially for graph neural networks. It

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

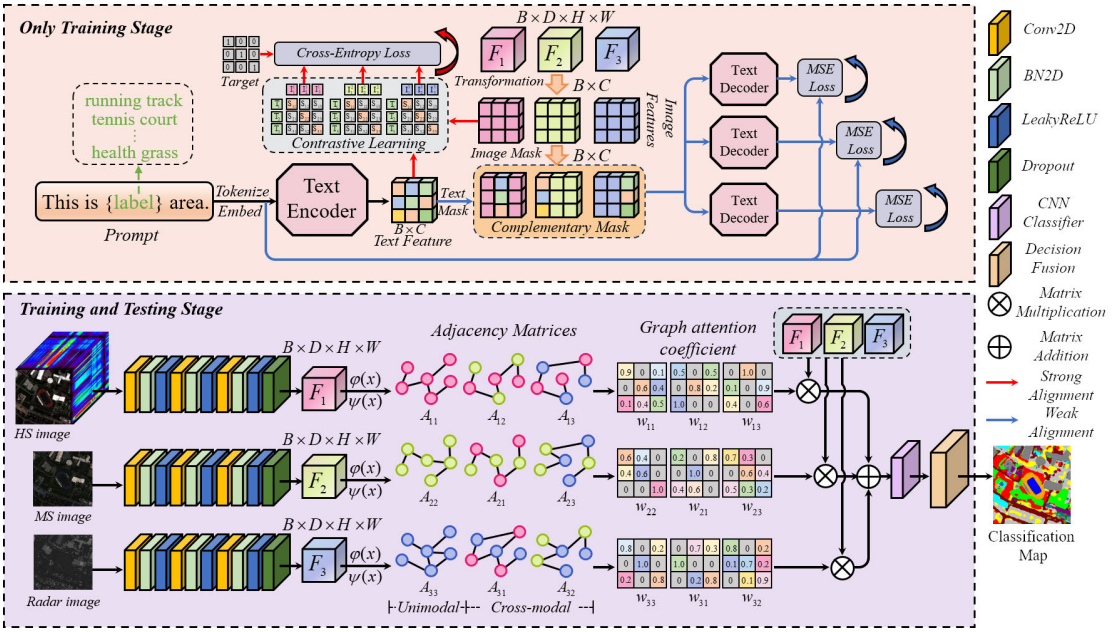


Fig. 1. Overall architecture of the proposed LKA-GFNet.

aims to explicitly model relationships between nodes to reveal latent structural information in the data, thereby supporting subsequent operations such as graph convolution, graph attention, or graph matching. In RS, it is common to treat pixels, superpixels, or objects as nodes and define topological relationships among them, forming a structured representation for mining spatial context and cross-modal dependencies. Most graph construction methods rely on spatial proximity or spectral similarity to reflect correlations among spatially adjacent or spectrally akin objects.

Wang et al. [38] proposed a HS change detection network based on GCNs, which groups HS pixels into superpixels as graph nodes, with edges constructed from superpixel adjacency. Du et al. [39] developed a multimodal graph for HS and LiDAR fusion classification by combining interpixel distance matrices from both modalities and determining edges using a k-nearest-neighbors (k-NNs) strategy. Yang et al. [40] designed a graph-based progressive hierarchical network for cross-scene RS image classification, treating all pixels in HS and LiDAR as adjacent nodes and constructing similarity matrices using spectral angle and Euclidean distance as edge weights. Chen and Wang [41] introduced a multiorder GCN for few-shot RS scene classification, building an adjacency matrix via k-NN and a radial basis function kernel to strengthen connections between nodes of the same category.

Although graph structures effectively model global spatial dependencies and provide a basis for GCN operations, most current graph construction strategies rely on fixed rules such as k-NN or preset thresholds. Such static approaches often struggle to adapt to complex land-cover distributions and may introduce noisy connections or miss meaningful long-range dependencies in heterogeneous data. Therefore, developing adaptive graph construction mechanisms that balance robustness to noise and structural informativeness remains an important research direction.

III. Methodology

A. Overall Architecture

The overall framework of the proposed LKA-GFNet is illustrated in Fig. 1. The model operates in two phases: training and testing. The training phase incorporates both a label-text prior branch and a tri-source image branch, while the testing phase includes only the tri-source image branch, as label information is available only during training. During training, textual semantic priors and image features are jointly optimized to achieve cross-modal alignment. Under supervised loss constraints, image features are driven to converge into a unified semantic space, enhancing both cross-modal consistency and discriminability. During testing, the learned semantically enriched representations are utilized for fusion classification of input tri-source images, maintaining robust performance even without explicit textual guidance. Specifically, in the text branch, we construct textual prompts for different land-cover categories using the template: “This is label area.”, where label is replaced by the actual category name. These prompts are fed into the proposed PSAA module to generate high-dimensional semantic embeddings. This module treats the textual priors as unified semantic anchors and employs a dual-stage (strong-weak) alignment mechanism to progressively drive the tri-source image features F_1, F_2, and F_3 to converge into a shared semantic space. Meanwhile, in the image branch, the aligned tri-source image features are input into the proposed CT-GDF module. This module employs a compound thresholding strategy to construct multilevel graph structures within and across modalities, explicitly modeling feature correlations and disparities. The final classification result is obtained through a classifier and decision fusion.

The PSAA and CT-GDF modules are not independent but exhibit a clear cascaded dependency. PSAA combines strong contrastive alignment with weak cross-modal

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

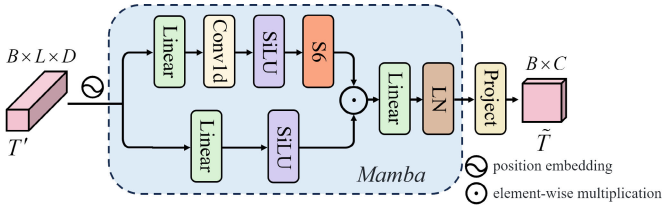


Fig. 2. Workflow of the text encoder.

reconstruction to shape a more consistent semantic geometry for tri-source features in the shared embedding space, which makes cross-modal distances more comparable and stable. CT-GDF then constructs and refines the graph structure based on this distance matrix, and its candidate neighbor selection and threshold-based edge pruning directly depend on whether the distance distribution can reflect true semantic proximity. Furthermore, the two-stage threshold mechanism in CT-GDF refines node connections from coarse to fine. This refinement suppresses noisy connections introduced by residual alignment errors and reduces the risk of error propagation. Therefore, the two modules complement each other and provide synergistic gains for tri-source fusion.

B. PSAA Module

Multisource RS data often exhibit significant modality discrepancies and semantic gaps due to differences in imaging mechanisms, spatial resolution, and information representation. Straightforward cross-modal feature fusion can lead to distribution shifts and semantic inconsistencies. To address this, we propose the PSAA module, which leverages geoscience textual priors to establish a unified semantic reference. It employs a dual-stage alignment strategy (strong and weak) to gradually steer tri-source image features into a shared semantic space, ensuring global consistency across modalities while preserving local discriminative representations.

Specifically, label-based textual prompts are first tokenized using the CLIP tokenizer [33], converting raw text into a numerical token sequence $T \in \mathbb{R}^{B\times L}$, where $B = 64$ denotes the batch size and $L = 77$ represents the sequence length. The tokenized input is embedded into the textual embedding space to yield a text feature $T' \in \mathbb{R}^{B\times L\times D}$, with an embedding dimension of $D = 64$. Subsequently, the text feature T' is fed into a Mamba-based text encoder, whose architecture is illustrated in Fig. 2. Built on state space models, Mamba leverages a selective mechanism to model sequences in linear time, capturing long-range dependencies with markedly lower computation and memory costs. This makes the encoder efficient and scalable for variable-length, long-text inputs, while achieving performance comparable to or better than transformers. Specifically, the text encoder first augments T' with learnable positional encodings $p \in \mathbb{R}^{B\times L\times D}$ to model token order and contextual dependencies. The resulting features are then fed into a Mamba block, which consists of a main branch, a gating branch, and a fusion stage. The main branch comprises a linear layer (128 dims), a 1-D convolutional layer (128 kernels, size 1×4), SiLU activation, and the S6 model for selective scanning. The gating branch has a linear layer (128 dims) followed by SiLU. The two branch outputs are combined by element-wise multiplication and further transformed by a linear layer (64 dims) followed by LayerNorm, producing $T'' \in \mathbb{R}^{B\times L\times D}$. Following CLIP, we take the hidden state at the EOS position from T'' as the sentence-level text representation and project it using a learnable matrix $W_{proj} \in \mathbb{R}^{D\times C}$ to obtain $\tilde{T} \in \mathbb{R}^{B\times C}$, where $C = 128$ is the projection dimension

$$\begin{cases} p_i = \underset{j}{argmax}\, T_{ij} \\ \tilde{T}_i = T''_{i,p_i} W_{proj}. \end{cases} \tag{1}$$

Meanwhile, three CNNs with identical architectures but independent weights are applied to the tri-source images, respectively, to extract heterogeneous image features $F_1, F_2, F_3 \in \mathbb{R}^{B\times D\times H\times W}$, where $D = 64$ denotes the feature dimension, and H and W represent the height and width of the images. The architecture of each CNN is shown in Fig. 1, consisting of three convolutional modules followed by a Dropout layer with a rate of 0.5. Each convolutional module is composed of a Conv2d layer (64 kernels, size 3×3), a BatchNorm2d layer, and a LeakyReLU activation function. To match the dimensionality of the text features, the image features F_1, F_2, F_3 first undergo global average pooling (GAP) to compress their spatial dimensions and are then passed through a Conv2d layer (C kernels, size 1×1) to adjust the channel dimension, thereby obtaining the features $\tilde{F}_1, \tilde{F}_2, \tilde{F}_3 \in \mathbb{R}^{B\times C}$. The dual-stage alignment is applied between $\tilde{T}$ and each image feature.

1) Strong Alignment (SA): Contrastive learning is used to align image and text features by maximizing similarity for positive sample pairs and minimizing it for negative pairs. We develop an SA strategy based on contrastive learning: Each image feature $\tilde{F} \in \{\tilde{F}_1, \tilde{F}_2, \tilde{F}_3\}$ is paired with its corresponding text embedding as a positive sample, and text embeddings from other classes as negatives. The cosine similarity $S \in \mathbb{R}^{B\times B}$ is computed as

$$S\left(\tilde{F}_i, \tilde{T}_j\right) = \frac{\sum_{k=1}^{C} \tilde{F}_{ik}\tilde{T}_{jk}}{\sqrt{\sum_{k=1}^{C} \tilde{F}_{ik}^2} \cdot \sqrt{\sum_{k=1}^{C} \tilde{T}_{jk}^2}}\, i, \quad j = 1, \ldots, B. \tag{2}$$

Subsequently, the image-to-text contrastive loss is

$$\mathscr{L}_{\mathrm{i2t}}\left(\tilde{F}_i, \tilde{T}_j\right) = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp\left(S_{ii}/\tau\right)}{\sum_{j=1}^{B} \exp\left(S_{ij}/\tau\right)} \tag{3}$$

where τ is a learnable temperature parameter. This objective explicitly enhances the similarity between image features and semantically matching textual descriptions, while reducing their similarity to descriptions of other categories. Such an SA strategy encourages image representations to conform more closely to geoscience concepts, thereby mitigating challenges such as high intraclass variance and interclass similarity. As a result, the model's final classification decisions not only rely on the image features themselves but also benefit from prior knowledge embedded within label text. Similarly, the text-to-image loss is computed

$$\mathscr{L}_{\mathrm{t2i}}\left(\tilde{T}_j, \tilde{F}_i\right) = -\frac{1}{B}\sum_{j=1}^{B} \log \frac{\exp\left(S_{jj}/\tau\right)}{\sum_{i=1}^{B} \exp\left(S_{ij}/\tau\right)}. \tag{4}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

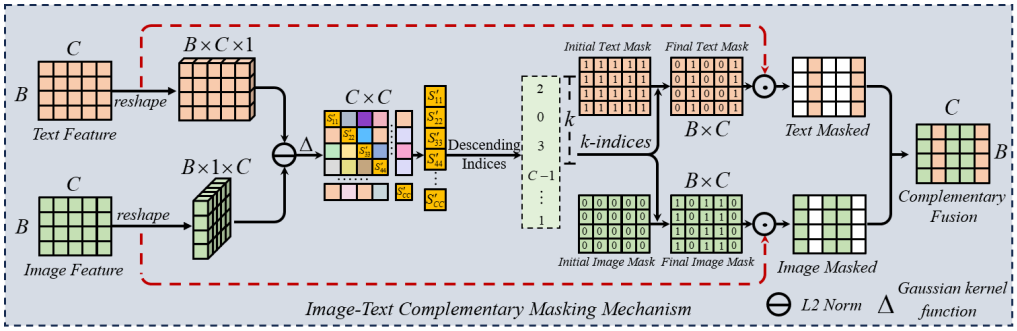


Fig. 3. Schematic of the image–text complementary masking mechanism.

Finally, the total contrastive loss is computed

$$\mathscr{L} = \frac{1}{2}\left(\mathscr{L}_{\mathrm{i2t}} + \mathscr{L}_{\mathrm{t2i}}\right). \tag{5}$$

Therefore, through the aforementioned process, the contrastive losses between the tri-source image features and the text feature, denoted as $\mathscr{L}_1$, $\mathscr{L}_2$, and $\mathscr{L}_3$, are obtained separately. The final SA loss is defined as

$$\mathscr{L}_{s_a} = \frac{1}{3}\left(\mathscr{L}_1 + \mathscr{L}_2 + \mathscr{L}_3\right). \tag{6}$$

2) Weak Alignment (WA): In contrast to SA, which imposes direct constraints on image–text correspondence in a global semantic space, WA aims to promote flexible cross-modal consistency through relative feature reconstruction. Inspired by the audio–video complementary fusion in [42], we design a WA strategy based on an image–text complementary masking mechanism. We formulate WA as a cross-modal masked semantic completion task. The core idea is that under the strong heterogeneity of tri-source RS data, directly enforcing global semantic-space consistency may over-constrain modality-specific information, thereby causing training instability. To address this, we select dimension-wise semantic channels that exhibit high consistency between image and text in a shared embedding space, mask these channels on the text side, and then complement and fuse them using the visible channels from the image side. Finally, the completed text features are fed into a decoder to reconstruct the original text. This introduces an explicit training constraint: only when the image representation carries semantic information consistent with the text can it reconstruct the masked textual semantics. Moreover, since the textual semantic prototype is unique, this process encourages tri-source heterogeneous image representations to move toward the same textual semantic prototype, thereby achieving semantic alignment across the heterogeneous tri-source data.

As illustrated in Fig. 3, the method first computes the Euclidean distance between each of the tri-source image features $\tilde{F} \in \{\tilde{F}_1, \tilde{F}_2, \tilde{F}_3\}$ and text feature $\tilde{T}$. This distance is converted into a similarity score $S' \in \mathbb{R}^{C\times C}$ using a Gaussian kernel

$$S'\left(\tilde{F}_i, \tilde{T}_j\right) = \exp\left(-\frac{\sum_{k=1}^{B}\left(\tilde{F}_{ki} - \tilde{T}_{kj}\right)^2}{2\sigma^2}\right) \quad i,\quad j = 1,\ldots,C \tag{7}$$

where σ is the bandwidth parameter of the Gaussian kernel. The diagonal elements $\{S'_{ii} | i = 1, 2, \ldots, C\}$ denote the degree of similarity between the ith feature dimensions of the image and text. The Top-k algorithm is then applied to select the indices corresponding to the k highest similarity values, where $k = \lfloor \rho \cdot C \rfloor$ and $\rho \in (0, 1)$ is a selection ratio. A complementary masking strategy is applied: In the text feature $\tilde{T}$, the dimensions at the selected indices are masked, while in the image feature $\tilde{F}$, only those indices are retained—all others are masked. The visible portions of the image and text features are then fused according to their indices to reconstruct a new text feature. Selecting the Top-k high-similarity feature dimensions for masking, rather than low-similarity dimensions, is based on the rationale that high-similarity dimensions are more likely to belong to the shared semantic feature subspace between images and text. This yields more reliable supervision and smoother gradients, making the cross-modal completion objective easier to optimize. In contrast, low-similarity dimensions tend to contain more modality-specific discrepancies and noise, and forcing their reconstruction can easily induce semantic drift and undermine alignment stability. The mask ratio ρ is used to balance alignment strength and semantic stability. When ρ is large, more text dimensions are masked, and the model must rely more on image features to complete the missing semantics, thereby strengthening the cross-modal consistency constraint. When ρ is small, a larger portion of the text remains visible, making the semantic prototype more stable, but the driving force that pulls image representations toward text semantics is relatively weakened. Considering the pronounced modality gap in tri-source RS data, we use a relatively large $\rho = 0.8$ to strengthen semantic injection while keeping part of the text dimensions as anchors to avoid drift. We further analyze the impact of different ρ values on classification accuracy in the parameter experiment section.

This complementary masking process is repeated for each of the three sources, producing reconstructed text features $\tilde{T}_{F_1}$, $\tilde{T}_{F_2}$, and $\tilde{T}_{F_3}$. Then, these features are, respectively, fed into the text decoders to reconstruct the original text information. The decoders mirror the encoder architecture, with Mamba as the backbone. Specifically, we first apply a linear layer ($L \times D$ dims) followed by reshaping to map $\tilde{T}_{F_1}/\tilde{T}_{F_2}/\tilde{T}_{F_3}$ back to the embedding space $\tilde{T}'_{F_1}/\tilde{T}'_{F_2}/\tilde{T}'_{F_3} \in \mathbb{R}^{B\times L\times D}$. The resulting feature $\tilde{T}'_{F_1}/\tilde{T}'_{F_2}/\tilde{T}'_{F_3}$ is then fed into a Mamba block to produce the reconstructed text features $\tilde{T}_1/\tilde{T}_2/\tilde{T}_3$. This Mamba block

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

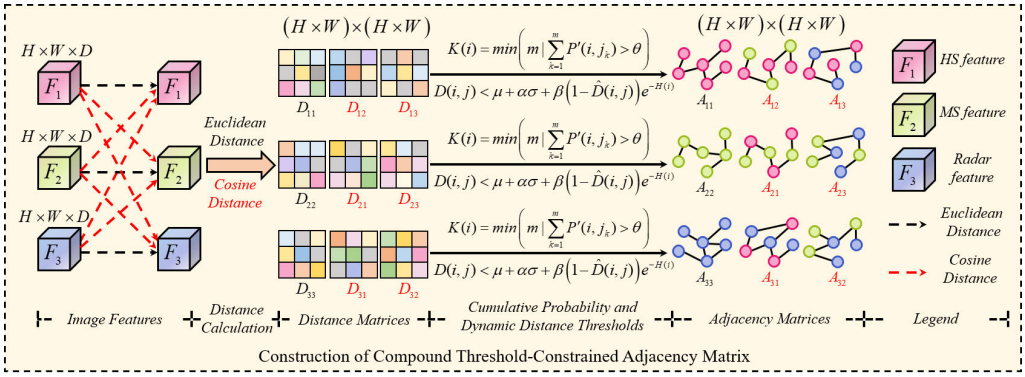


Fig. 4. Construction process of the adjacency matrices.

adopts the same architecture as that in the text encoder (see Fig. 2). The mean squared error (mse) loss is computed between each decoded feature and the original textual feature T'

$$\mathscr{L}_{mse}^{k} = mse\left(T', \tilde{T}_k\right), \quad k = 1, 2, 3. \tag{8}$$

This reconstruction process ensures that the recovered text preserves the original semantic information, enabling the image features to complement and reflect textual semantics. The WA strategy essentially constructs a semantic reasoning pathway from the tri-source images to a shared textual prototype. By implicitly establishing reasoning dependencies, it drives the image features to align spontaneously within a common semantic space. The total WA loss is defined as

$$\mathscr{L}_{w_a} = \frac{1}{3}\left(\mathscr{L}_{mse}^{1} + \mathscr{L}_{mse}^{2} + \mathscr{L}_{mse}^{3}\right). \tag{9}$$

C. CT-GDF Module

The CT-GDF module is designed to capture deep interactive relationships among tri-source heterogeneous features and support robust cross-modal decision-making within graph structures. Conventional graph fusion methods often rely on a single similarity metric or a fixed threshold to establish adjacency relationships, which can lead to redundant connections or omission of critical edges when handling multimodal features with distribution discrepancies and ambiguous class boundaries. To address this, we propose a compound threshold-constrained mechanism that adaptively constructs graph structures at both intramodal and cross-modal levels using cumulative probability and dynamic distance. We further employ graph attention to model deep intramodal and intermodal interactions.

The pixels in image features F_1, F_2, and F_3 are flattened and treated as nodes. Intramodal and cross-modal adjacency matrices are constructed to represent node relationships within and across modalities, respectively. As shown in Fig. 4, the intramodal adjacency matrices are built by computing the Euclidean distance between nodes within each modality

$$D_{nn}(i, j) = \sqrt{\sum_{k=1}^{C}\left(F_n^{ik} - F_n^{jk}\right)^2}, \quad n = 1, 2, 3 \tag{10}$$

where i and j denote nodes. For cross-modal adjacency, cosine distance is used to measure semantic relevance between nodes from different modalities

$$D_{pq}(i, j) = 1 - \frac{\sum_{k=1}^{C} F_p^{ik} F_q^{jk}}{\sqrt{\sum_{k=1}^{C}\left(F_p^{ik}\right)^2}\sqrt{\sum_{k=1}^{C}\left(F_q^{jk}\right)^2}}, p \quad q = 1, 2, 3; p \neq q. \tag{11}$$

We employ Euclidean distance within each modality and cosine distance across modalities for two reasons. First, intramodal features share a consistent measurement space, and Euclidean distance—being sensitive to magnitude differences—provides a reasonable measure of within-modality similarity. Second, intermodal features naturally differ in units and dynamic ranges, so directly applying Euclidean distance can be dominated by scale effects. We therefore adopt cosine distance, which is more robust to scaling, to describe cross-modal semantic neighborhood relations. Notably, our distance matrices are only used to construct the final binary (0/1) adjacency matrices. Edge weights are determined by subsequent graph-attention coefficients rather than the distances themselves. Moreover, during adjacency construction, the cumulative probability threshold converts distances into relative probabilities via Softmax, ensuring that edge selection is based on relative affinity rather than absolute scale. In the subsequent dynamic distance threshold, we introduce a node-wise normalized distance $\hat{D} \in [0, 1]$ to improve the comparability of local distance distributions across nodes, preventing any single distance metric from dominating the fusion process.

We propose a two-stage compound thresholding strategy to construct robust adjacency matrices.

1) Cumulative Probability Threshold: In the distance matrix D, for each node, its distances to all other nodes are sorted in ascending order to form a sorted index vector *Index*. The distance matrix D is converted into a probability matrix P via Softmax

$$P(i, j) = \frac{\exp\left(-D(i, j)/\eta\right)}{\sum_{k=1}^{N}\exp\left(-D(i, k)/\eta\right)} \tag{12}$$

where η is the temperature hyperparameter and N denotes the number of image nodes. The probability matrix P is then rearranged according to the index vector *Index*, yielding the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

sorted probability matrix P'. In P', nodes are arranged in descending order of relevance to the reference node. For each node i, the cumulative probability is computed sequentially over the ordered nodes in P' until it exceeds a predefined threshold θ. The number of nodes m accumulated at this point is recorded, and the first m nodes in P' are selected as the candidate adjacent nodes for node i. In the adjacency matrix, the entries corresponding to the original sorted indices (from *Index*) of these m nodes are set to 1 and the remaining entries to 0, completing the initial adjacency construction. The cumulative probability threshold θ controls the coverage of the candidate neighborhood and determines an adaptive upper bound on the number of candidate neighbors for each node. We set $\theta = 0.8$ to prioritize neighborhood coverage and graph connectivity, preventing information propagation from being limited in sparse regions due to an overly small candidate set. Any weakly related edges introduced in this coarse stage are subsequently pruned by the second-stage dynamic distance threshold, resulting in a stable coarse-to-fine graph construction procedure. Our parameter analysis further shows that $\theta = 0.8$ yields the best classification accuracy. The cumulative probability threshold criterion can be described as follows:

$$K(i) = min\left(m | \sum_{k=1}^{m} P'(i, j_k) > \theta\right), \quad m \in \{1, 2, \ldots, N\} \tag{13}$$

where i denotes the current node, and j_k represents the sorted nodes.

2) Dynamic Distance Threshold: The cumulative probability threshold performs a coarse-grained selection of candidate adjacent nodes for each node. To further refine this set, we propose a dynamic distance threshold criterion that adaptively filters noisy connections by incorporating global statistics of the distance distribution and local entropy information. This approach mitigates issues of excessive sparsity or density caused by fixed thresholds, ensuring robust graph construction through a two-step refinement process—from global candidate selection to local fine-grained pruning. Specifically, the global statistics (mean and standard deviation) of the distance matrix are first computed

$$\begin{cases} \mu = \dfrac{1}{N^2} \displaystyle\sum_{i=1}^{N} \sum_{j=1}^{N} D(i, j) \\ \sigma = \sqrt{\dfrac{1}{N^2} \displaystyle\sum_{i=1}^{N} \sum_{j=1}^{N} (D(i, j) - \mu)^2}. \end{cases} \tag{14}$$

Then, for each node i, the probability-distribution entropy is calculated from the probability matrix P

$$H(i) = -\sum_{j=1}^{N} P(i, j) \log(P(i, j) + \varepsilon) \tag{15}$$

where $\varepsilon = 1e^{-8}$ is a smoothing term. A decay factor based on this entropy is constructed as follows:

$$decay(i, j) = \exp(-H(i)). \tag{16}$$

The dynamic threshold $\mathscr{T}(i, j)$ for a node pair (i, j) is defined as

$$\mathscr{T}(i, j) = \mu + \alpha\sigma + \beta\left(1 - \hat{D}(i, j)\right) \cdot decay(i, j). \tag{17}$$

A candidate edge between nodes i and j is retained only if $D(i, j) < \mathscr{T}(i, j)$; otherwise, it is discarded. In (17), $\mu + \alpha\sigma$ is the global threshold baseline, μ defines the overall distance center, σ characterizes the dispersion, and α controls the global sparsity. Larger α produces a denser graph, while smaller α results in a sparser one. We set $\alpha = 0.08$ because this value introduces a small yet effective standard-deviation-level offset relative to the global mean. Our sensitivity analysis further confirms that $\alpha = 0.08$ provides a favorable tradeoff between stability and performance across multiple datasets. $\hat{D}(i, j) = (D(i, j)/\max_k D(i, k))$ denotes the normalized distance, which converts edge distances into a node-level relative scale and thus improves the comparability of local distance distributions across nodes. The neighbor term $\beta(1 - \hat{D}(i, j))$, constructed based on the normalized distance, makes the threshold more inclined to retain closer candidate edges. Specifically, $\beta(1 - \hat{D}(i, j))$ increases as the distance decreases, which moderately relaxes the threshold and prioritizes reliable nearest neighbors. The parameter β controls the magnitude of the local adjustment, ensuring that the neighbor term only serves as a lightweight refinement to the global baseline. If β is too large, local distance fluctuations may dominate the threshold, destabilizing the graph structure and introducing weakly correlated edges. Therefore, we adopt a small value $\beta = 0.01$ so that the local term does not overly interfere with the global baseline, suppressing noisy edges while improving the retention of nearest neighbors. Our parameter sensitivity experiments also verify that this setting achieves the best classification performance. $decay(i, j)$ is an entropy-decay factor constructed from the probability-distribution entropy $H(i)$, capturing uncertainty in the local neighborhood structure. A larger $H(i)$ implies a more diffuse neighborhood probability distribution, suggesting that the node may lie near class boundaries or be corrupted by noise. In this case, $decay(i, j)$ decreases and the dynamic threshold $\mathscr{T}(i, j)$ becomes smaller, leading to a more conservative edge-keeping policy that reduces noisy links and limits error propagation. When $H(i)$ is small, only a few neighbors dominate the distribution and the neighborhood structure is clearer, so $decay(i, j)$ and $\mathscr{T}(i, j)$ increase to retain more reliable nearest-neighbor connections. In this way, the entropy-decay factor acts as an uncertainty-aware gate: uncertain nodes adopt conservative connectivity, whereas confident nodes can leverage neighborhood structure more fully.

The final adjacency matrix integrates both the cumulative probability and dynamic distance criteria

$$A(i, j) = \begin{cases} 1, & if\ min\left(m | \sum_{k=1}^{m} P'(i, j_k) > \theta\right) \,\&\, (D(i, j) < \mathscr{T}(i, j)) \\ 0, & otherwise. \end{cases} \tag{18}$$

The proposed two-stage thresholding strategy is to adapt to local variations in sample density while suppressing noisy

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

edges. In RS scenes, samples are often distributed with nonuniform density: some nodes are surrounded by dense neighborhoods, whereas others lie in sparse areas. Using a fixed neighbor rule (e.g., k-NN or a single distance cutoff) tends to connect sparse-region nodes to distant and noisy neighbors, while potentially discarding many informative neighbors in dense regions. Therefore, we design a two-stage thresholding strategy to dynamically adjust neighborhood size and filtering strength, yielding a more stable graph structure. First, a cumulative probability threshold uses probability-mass coverage instead of a fixed k, yielding smaller candidate neighborhoods in dense regions and larger candidate sets in sparse regions, which avoids graphs that are overly dense or overly sparse. Second, we apply a dynamic distance threshold for fine-grained denoising within the candidate set. This threshold is anchored by a global baseline defined by the mean and standard deviation of the distance distribution, and is further refined by a nearest-neighbor term and entropy-based gating that enforces more conservative pruning for uncertain nodes. Overall, the coarse-to-fine thresholding yields a more robust adjacency structure under varying density and noise levels, and is more stable than graph construction with a single distance threshold.

After obtaining the intramodal and cross-modal adjacency matrices, we combine them with an attention mechanism to form graph-attention coefficients

$$w_{pq} = \left\{ Softmax\left(A_{pq} \odot \left(F_p F_q^T\right)\right) | p, q = 1, 2, 3 \right\} \tag{19}$$

where $\odot$ denotes element-wise multiplication. This formulation ensures that only attention coefficients between connected nodes are retained, thereby preventing irrelevant nodes from contributing to information aggregation. The attention matrix w_{pq} can be categorized into two types: the intramodal graph-attention coefficients $\{w_{pq} | p = q\}$ and the intermodal graph-attention coefficients $\{w_{pq} | p \neq q\}$. To integrate node information within and across modalities, the two types of graph-attention coefficients are combined as follows: w_{11}, w_{12}, and w_{13} are concatenated to form $W_1 \in \mathbb{R}^{N \times 3N}$; similarly, w_{21}, w_{22}, and w_{23} form $W_2 \in \mathbb{R}^{N \times 3N}$; and w_{31}, w_{32}, and w_{33} form $W_3 \in \mathbb{R}^{N \times 3N}$

$$\begin{cases} W_1 = [w_{11} \| w_{12} \| w_{13}] \\ W_2 = [w_{21} \| w_{22} \| w_{23}] \\ W_3 = [w_{31} \| w_{32} \| w_{33}] \end{cases} \tag{20}$$

where $[\cdot]$ denotes feature concatenation. The tri-source heterogeneous features F_1, F_2, and F_3 are flattened and concatenated to form the feature $F_{cat} \in \mathbb{R}^{3N \times D}$. Using the fused graph-attention matrices W_1, W_2, and W_3, the node information is aggregated to generate deeply interactive features $\hat{F}_1$, $\hat{F}_2$, and $\hat{F}_3$

$$\begin{cases} \hat{F}_1^{ij} = \sum_{k=1}^{3N} W_1^{ik} F_{cat}^{kj} \\ \hat{F}_2^{ij} = \sum_{k=1}^{3N} W_2^{ik} F_{cat}^{kj} \\ \hat{F}_3^{ij} = \sum_{k=1}^{3N} W_3^{ik} F_{cat}^{kj}. \end{cases} \tag{21}$$

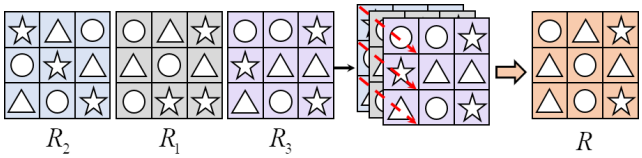


Fig. 5. Decision fusion strategy.

The final fused feature $\hat{F}$ is obtained through

$$\hat{F} = \hat{F}_1 + \hat{F}_2 + \hat{F}_3. \tag{22}$$

The feature $\hat{F}$ is then passed into a convolutional classifier to obtain the final prediction. The classifier consists of two stages: the first includes Conv2d, BatchNorm, and LeakyReLU, and the second comprises GAP, Conv2d, and Softmax. GAP reduces spatial dimensions, and both convolutional layers use 1×1 kernels. The final layer maps the channel dimension to the number of classes.

To further enhance the discriminative capability of the network, this article designs a decision fusion strategy. The tri-source image features F_1, F_2, and F_3 are processed through graph-attention fusion to obtain a refined classification result R_1. Meanwhile, max-pooling is applied to F_1, F_2, and F_3 to extract discriminative features F_1', F_2', and F_3', preserving salient characteristics such as edges and textures with improved noise robustness. These features are similarly processed through graph-attention fusion to yield a discriminative classification result R_2. In addition, element-wise summation of F_1, F_2, and F_3 results in a direct classification output R_3.

The decision fusion rule is defined as: if a pixel receives the same label in at least two of the three results, that label is assigned; if all three results disagree, the pixel is classified according to R_1. This decision fusion strategy is not a simple ensemble of multiple classifiers. Instead, it constructs complementary discriminative evidence tailored to the error patterns in tri-source RS fusion. R_1 relies on graph-attention fusion to impose structural consistency, offering the strongest semantic reasoning path and mitigating semantic mismatches induced by cross-modal shifts. R_2 uses max-pooling to emphasize salient local textures and boundaries, making it more robust to noise and local anomalies. R_3 adopts direct fusion as a stable baseline to avoid over-reliance on any single mechanism. We apply majority voting to reduce sporadic errors from individual paths, and when all three predictions conflict, we fall back to R_1 to control risk on difficult samples due to its structure-aware reasoning. The process is illustrated in Fig. 5.

D. Training Strategy

The proposed LKA-GFNet extracts a patch of size $r \times r$ centered on each training sample and generates the corresponding textual prompt based on the point's class label. The training set is defined as: $\{(X_{HS}^i, X_{MS}^i, X_{Radar}^i, T^i), Y^i\}_{i=1}^U$, where $(X_{HS}^i, X_{MS}^i, X_{Radar}^i)$ denotes the HS, MS, and radar image patches corresponding to the ith training point, T^i represents the textual prompt, Y^i is the class label, and U indicates the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

total number of training points. The model is optimized using the cross-entropy loss function

$$\mathcal{L}_{CE} = -\frac{1}{U}\sum_{u=0}^{U-1}\sum_{m=0}^{M-1} y_{u,m} log\left(p_{u,m}\right) \tag{23}$$

where M denotes the number of classes, $p_{u,m}$ represents the predicted probability for class m, and $y_{u,m}$ is the ground truth label.

Algorithm 1 Overall Procedure of LKA-GFNet

Require: Tri-source RS images HS, MS, and $Radar$; Label-text prompt; Training samples; Test samples;

Ensure: Classification prediction R;

1: All parameters in LKA-GFNet are initialized, and the training set $\{(X^i_{HS}, X^i_{MS}, X^i_{Radar}, T^i), Y^i\}^U_{i=1}$ and test set $\{(X^{i'}_{HS}, X^{i'}_{MS}, X^{i'}_{Radar})\}^{U'}_{i'=1}$ are generated.

2: **while** $epoch < epochs$, **do:**

step 1: Features F_1, F_2, and F_3 are obtained by feeding X^i_{HS}, X^i_{MS}, and X^i_{Radar} into their corresponding image branches, respectively, while textual feature $\tilde{T}$ is extracted by inputting T^i into the text branch. The SA and WA losses between the tri-source features and the text are then computed according to Equations (6) and (9);

Step 2: The tri-source heterogeneous features F_1, F_2, and F_3 are fed into the proposed CT-GDF module to generate the final classification result;

step 3: The overall loss $\mathcal{L}_{total}$ of the network is calculated according to Equation (24), and all parameters of LKA-GFNet are updated via backpropagation;

3: The test set is input into LKA-GFNet, yielding the final classification prediction results.

Since the decision fusion strategy generates three probability predictions, denoted as p_1, p_2, and p_3, the cross-entropy loss is computed individually for each output, yielding $\mathcal{L}^1_{CE}$, $\mathcal{L}^2_{CE}$, and $\mathcal{L}^3_{CE}$. The total loss integrates these with the strong and WA losses from (6) and (9)

$$\mathcal{L}_{total} = \mathcal{L}^1_{CE} + \mathcal{L}^2_{CE} + \mathcal{L}^3_{CE} + \mathcal{L}_{s_a} + \mathcal{L}_{w_a}. \tag{24}$$

The overall procedure of LKA-GFNet is summarized in Algorithm 1.

IV. Experiments

A. Experimental Setup and Training Details

The proposed LKA-GFNet is implemented in the PyTorch framework and runs on a computer equipped with Ubuntu 20.04 and an NVIDIA GeForce RTX 4080 GPU. The batch size is set to 64. During training, the random seed is fixed at 40, and the network is trained for 500 epochs using the Adam optimizer with an initial learning rate of $1e^{-4}$. To improve training stability, a StepLR learning-rate schedule is adopted: the learning rate is decayed by a factor of $\gamma = 0.9$ every 50 epochs, and the learning rate is updated at the end of each epoch. All modules in LKA-GFNet use PyTorch default parameter initialization, and the cross-entropy loss is used as the supervised objective.

In the data preprocessing stage, min–max normalization is applied separately to each modality to eliminate scale variations across different sensor data. This study does not employ data augmentation techniques such as random flipping or rotation, in order to objectively evaluate the model's intrinsic ability to learn and fuse multisource features without introducing augmentation-induced biases.

B. Tri-Source Datasets

The effectiveness of LKA-GFNet is evaluated on four tri-source RS datasets: Houston2013 (HS+MS+LiDAR), Augsburg City (HS+MS+SAR), Beijing (HS+MS+SAR), and Wuhan (HS+MS+SAR). Table I summarizes the key characteristics of each dataset, and Table II reports the numbers of training and testing samples. We adopt the official training/testing split for Houston2013. For the other three datasets without official splits, we randomly select 200 labeled pixels per class for training. Given their excessively large labeled sets, we randomly select 1% of the remaining labeled pixels per class for testing to keep evaluation computationally feasible.

The four tri-source datasets offer complementary coverage in sensor combinations, scene types, and class sample scales, making them representative of typical challenges in practical tri-source RS classification. Common modalities in multisource fusion include HS, MS, LiDAR, and SAR, all of which are covered by our datasets. Specifically, Houston2013 uses HS+MS+LiDAR, highlighting the need to fuse spectral cues with spatial context and geometric structure. The other datasets use HS+MS+SAR, introducing SAR whose imaging mechanism differs substantially from optical sensors, leading to stronger cross-modal distribution gaps and better reflecting practical heterogeneous optical-radar fusion scenarios. Moreover, Houston2013 corresponds to fine-grained urban classification, where many classes have similar appearances and are prone to interclass confusion. The Augsburg, Beijing, and Wuhan datasets cover mixed urban and suburban regions, and their classes are mostly coarse-grained and often exhibit larger intraclass spectral variability. Meanwhile, these datasets exhibit pronounced class imbalance, requiring models to balance fine-grained discrimination under limited supervision with generalization to complex scenes.

C. Parameter Settings

1) *Mask Ratio ρ in the Image–Text Masking Mechanism:* In the PSAA module, we design a WA strategy based on an image–text masking mechanism. The mask ratio ρ controls the proportion of feature dimensions to be masked. This ratio influences the degree of feature obfuscation and thus the cross-modal complementary learning. A too-small value may fail to encourage robust alignment, while too large a value could degrade semantic integrity. As illustrated in Fig. 6, the model achieves optimal classification accuracy when $\rho = 0.8$.

2) *Cumulative Probability Threshold θ:*
In the CT-GDF module, a dual-threshold strategy is used for adjacency matrix construction. The cumulative probability threshold θ is primarily used for the initial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

TABLE I

DETAILED CHARACTERISTICS OF THE FOUR TRI-SOURCE DATASETS

Datasets	Houston2013 [9][43]			Augsburg City [30]			Beijing [30]			Wuhan [30]		
Region	Houston, Texas, USA			Augsburg, Germany			Beijing, China			Wuhan, Hubei, China		
Sensor Type	HS	MS	LiDAR	HS	MS	SAR	HS	MS	SAR	HS	MS	SAR
Image Size	349×1905	349×1905	349×1905	886×1360	886×1360	886×1360	1000×1000	1000×1000	1000×1000	1100×1200	1100×1200	1100×1200
Number of Bands	144	8	1	242	4	2	116	4	2	116	4	2
Spatial Resolution	10m	2.5m	2.5m	10m	10m	10m	10m	10m	10m	10m	10m	10m
Wavelength Range	380-1050nm	/	/	400-2500nm	/	/	400-2500nm	/	/	400-2500nm	/	/
Collection source	CASI-1500	/	/	HyMap/HySpex	Sentinel-2	Sentinel-1	AHSI	GF-6	GF-3	GF-5	GF-6	GF-3

TABLE II

NUMBER OF TRAINING AND TESTING SAMPLES PER CLASS ACROSS THE FOUR DATASETS

No.	Class name	Houston2013		No.	Class name	Augsburg City		Beijing		Wuhan	
		Train	Test			Train	Test	Train	Test	Train	Test
1	Health grass	198	1053	1	Surface water	200	215	200	510	200	3620
2	Stressed grass	190	1064	2	Street network	200	1359	200	1086	200	3380
3	Synthetic grass	192	505	3	Urban fabric	200	2754	200	3546	200	2293
4	Trees	188	1056	4	Industrial, commercial, and transport	200	1504	200	427	200	2116
5	Soil	186	1056	5	Mine, dump, and construction sites	200	171	200	641	200	61
6	Water	182	143	6	Artificial vegetated areas	200	1001	200	1425	200	1107
7	Residential	196	1072	7	Arable land	200	2770	200	461	200	121
8	Commercial	191	1053	8	Permanent crops	200	26	200	157	200	120
9	Road	193	1059	9	Pastures	200	494	200	18	200	112
10	Highway	191	1036	10	Forests	200	1193	200	1633	200	141
11	Railway	181	1054	11	Shrub	200	438	200	60	200	372
12	Parking lot 1	192	1041	12	Open spaces with no vegetation	200	10	200	12	200	108
13	Parking lot 2	184	285	13	Inland wetlands	200	46	200	5	200	128
14	Tennis court	181	247	/	/	/	/	/	/	/	/
15	Running track	187	473	/	/	/	/	/	/	/	/
	Total	2832	12197		Total	2600	11981	2600	9981	2600	13679

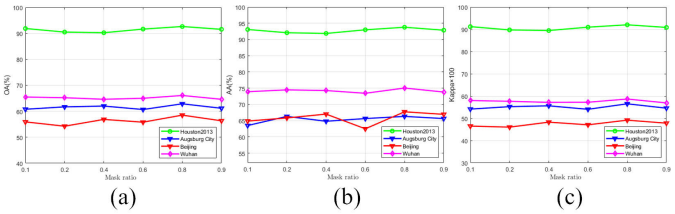


Fig. 6. Classification accuracy under different mask ratios ρ. (a) OA. (b) AA. (c) Kappa.

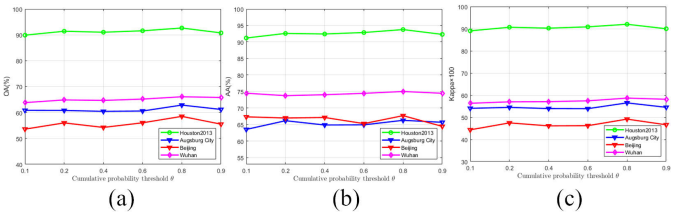


Fig. 7. Classification accuracy under different cumulative probability thresholds θ. (a) OA. (b) AA. (c) Kappa.

screening of candidate node connections. The value of θ influences the coverage of candidate adjacent nodes: a lower threshold results in a sparser adjacency set with only the most relevant nodes, while a higher threshold retains more connections—including potentially noisy edges. An appropriate θ is essential for balancing structural sparsity and informational completeness. Experimental results in Fig. 7 indicate that $\theta = 0.8$ delivers the best classification performance.

3) *Patch Size:* The patch size determines the extent of local spatial context available to the model for each sample and is a key hyperparameter affecting classification performance. A small patch may lose neighborhood information, leading to weak texture and boundary characterization, whereas an overly large patch tends to introduce mixed pixels and cross-class context, reducing discriminability. We conduct sensitivity experiments on four datasets with different patch sizes (7×7, 9×9, 11×11, 13×13, and 15×15) to identify an appropriate input scale. As shown in Fig. 8, Houston2013 achieves

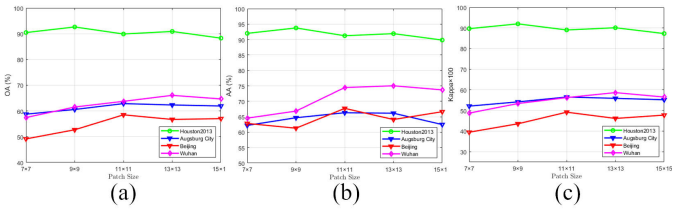


Fig. 8. Classification accuracy under different patch sizes. (a) OA. (b) AA. (c) Kappa.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

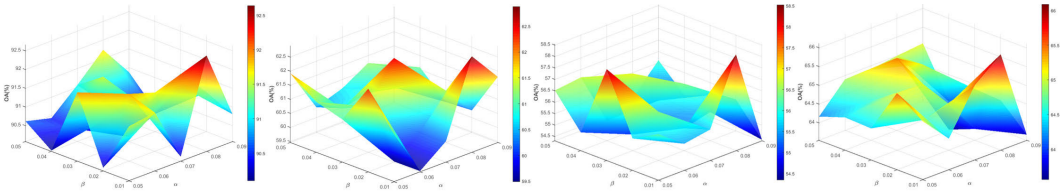


Fig. 9. Classification accuracy under different α and β values. (a) Houston2013. (b) Augsburg City. (c) Beijing. (d) Wuhan.

TABLE III
QUANTITATIVE COMPARISON OF DIFFERENT ALGORITHMS ON HOUSTON2013 DATASET

No.	CCR-Net	SepDGConv	AM3Net	ExViT	MFT	MSFMamba	Fusion-CNN	S2FL	MDAS	S3F2Net	DAHGMN	Proposed
1	82.10 ± 0.20	76.54 ± 6.18	78.07 ± 1.22	80.73 ± 1.75	**87.85 ± 8.05**	80.44 ± 0.40	80.77 ± 0.34	80.86 ± 0.33	81.82 ± 0.74	82.38 ± 0.47	81.67 ± 0.27	82.48 ± 0.20
2	84.77 ± 0.53	81.77 ± 3.46	79.31 ± 3.97	82.43 ± 0.80	83.74 ± 0.13	**99.48 ± 0.47**	74.25 ± 3.46	84.35 ± 0.20	83.79 ± 0.06	90.98 ± 8.24	84.82 ± 0.07	95.44 ± 2.46
3	97.43 ± 0.00	87.42 ± 13.58	92.02 ± 6.63	94.36 ± 1.54	96.64 ± 0.28	96.14 ± 0.42	**99.70 ± 0.14**	97.72 ± 0.42	97.62 ± 0.00	91.09 ± 0.84	92.68 ± 3.08	96.44 ± 0.00
4	89.49 ± 6.29	89.91 ± 4.89	68.81 ± 3.83	82.39 ± 1.48	92.81 ± 0.40	95.69 ± 0.47	85.28 ± 0.33	88.45 ± 2.28	87.69 ± 0.80	**97.73 ± 0.27**	91.24 ± 0.34	94.04 ± 3.49
5	98.72 ± 0.60	97.63 ± 0.81	98.72 ± 0.07	99.48 ± 0.34	99.86 ± 0.07	**100.00 ± 0.00**	98.86 ± 0.53	**100.00 ± 0.00**	98.77 ± 0.00	99.91 ± 0.13	99.96 ± 0.06	**100.00 ± 0.00**
6	95.80 ± 0.00	95.80 ± 0.00	84.12 ± 2.26	92.31 ± 4.94	89.51 ± 1.98	**100.00 ± 0.00**	97.55 ± 1.48	91.61 ± 11.87	97.55 ± 2.48	97.90 ± 2.97	94.76 ± 2.47	**100.00 ± 0.00**
7	81.48 ± 1.52	76.03 ± 8.84	62.60 ± 1.71	86.10 ± 2.11	83.63 ± 1.51	73.18 ± 0.99	88.34 ± 0.00	66.93 ± 0.99	79.99 ± 1.39	**93.19 ± 2.64**	88.34 ± 6.34	89.84 ± 0.93
8	94.40 ± 0.94	83.76 ± 5.91	86.24 ± 1.33	**95.02 ± 0.87**	87.56 ± 11.02	93.97 ± 0.34	80.20 ± 0.20	74.26 ± 0.94	77.02 ± 1.08	83.67 ± 8.06	83.77 ± 1.75	93.21 ± 3.56
9	87.20 ± 2.74	81.16 ± 7.14	58.52 ± 7.79	79.65 ± 1.94	87.78 ± 0.33	85.74 ± 1.47	76.58 ± 2.14	77.67 ± 2.47	77.81 ± 0.54	**92.30 ± 5.54**	91.65 ± 4.48	87.35 ± 2.40
10	68.10 ± 0.20	61.29 ± 7.23	63.60 ± 6.67	64.05 ± 3.49	65.44 ± 0.82	66.99 ± 0.54	42.62 ± 0.75	80.89 ± 4.50	66.12 ± 0.14	78.19 ± 3.14	64.58 ± 0.13	**82.68 ± 1.70**
11	93.93 ± 0.81	81.92 ± 1.01	91.11 ± 1.11	92.65 ± 1.41	94.55 ± 2.75	84.06 ± 2.28	78.99 ± 7.18	80.03 ± 2.75	**96.73 ± 1.95**	94.92 ± 3.15	89.80 ± 5.43	96.63 ± 1.00
12	96.83 ± 0.14	94.14 ± 4.34	97.83 ± 0.76	97.41 ± 0.68	95.30 ± 2.31	**99.04 ± 0.41**	75.55 ± 1.83	97.50 ± 0.41	96.30 ± 0.89	96.88 ± 2.38	98.18 ± 0.54	95.44 ± 1.70
13	75.44 ± 1.98	70.35 ± 5.70	73.41 ± 10.02	80.88 ± 4.72	74.57 ± 0.25	84.39 ± 0.25	67.19 ± 1.24	71.75 ± 2.23	85.27 ± 1.49	91.40 ± 0.25	91.06 ± 1.24	**91.58 ± 0.49**
14	96.76 ± 4.01	**100.00 ± 0.00**	97.37 ± 3.72	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	92.91 ± 1.43	99.80 ± 0.29	98.38 ± 0.00	99.19 ± 1.15	99.80 ± 0.28	**100.00 ± 0.00**
15	94.50 ± 3.89	93.98 ± 6.43	95.70 ± 2.19	93.45 ± 8.37	99.58 ± 0.60	**100.00 ± 0.00**	98.31 ± 1.20	99.47 ± 0.15	96.41 ± 2.10	96.09 ± 4.34	**100.00 ± 0.00**	99.58 ± 0.60
OA (%)	88.37 ± 0.86	83.32 ± 2.78	79.88 ± 0.78	86.88 ± 0.21	88.64 ± 0.69	88.99 ± 0.01	80.16 ± 0.95	84.48 ± 0.13	86.06 ± 0.25	91.49 ± 0.19	88.56 ± 0.25	**92.49 ± 0.22**
AA (%)	89.13 ± 0.92	84.79 ± 3.13	81.83 ± 1.70	88.06 ± 0.46	89.25 ± 0.44	90.61 ± 0.01	82.48 ± 0.70	86.09 ± 0.68	88.09 ± 0.19	92.39 ± 0.47	90.16 ± 0.23	**93.65 ± 0.18**
Kappa×100	87.40 ± 0.93	81.96 ± 3.03	78.14 ± 0.78	85.76 ± 0.23	87.66 ± 0.75	88.04 ± 0.01	78.52 ± 1.03	83.14 ± 0.15	84.88 ± 0.27	90.78 ± 0.20	87.62 ± 0.25	**91.84 ± 0.24**

TABLE IV
QUANTITATIVE COMPARISON OF DIFFERENT ALGORITHMS ON AUGSBURG CITY DATASET

No.	CCR-Net	SepDGConv	AM3Net	ExViT	MFT	MSFMamba	Fusion-CNN	S2FL	MDAS	S3F2Net	DAHGMN	Proposed
1	58.60 ± 9.87	66.51 ± 5.26	53.99 ± 1.27	67.91 ± 5.26	65.82 ± 4.28	72.33 ± 0.33	62.56 ± 0.33	71.16 ± 1.31	**75.35 ± 0.66**	71.63 ± 0.66	75.12 ± 2.96	73.49 ± 0.00
2	21.19 ± 6.14	25.49 ± 9.21	11.60 ± 4.96	20.16 ± 7.81	14.57 ± 7.38	32.01 ± 0.31	13.17 ± 3.64	39.44 ± 0.62	**39.66 ± 1.46**	21.19 ± 8.53	29.47 ± 4.95	8.79 ± 8.27
3	62.82 ± 12.94	59.68 ± 13.02	68.24 ± 9.38	56.35 ± 6.26	**77.07 ± 11.02**	57.19 ± 0.16	72.33 ± 3.34	37.37 ± 0.87	50.78 ± 0.95	71.48 ± 7.68	59.55 ± 2.77	75.31 ± 12.84
4	48.01 ± 3.20	55.02 ± 2.21	63.33 ± 7.77	60.74 ± 18.85	54.06 ± 8.37	60.64 ± 0.09	52.06 ± 0.66	42.82 ± 0.19	60.24 ± 0.19	**72.24 ± 0.71**	52.99 ± 4.23	65.13 ± 8.51
5	39.18 ± 2.48	54.97 ± 0.82	65.22 ± 7.87	52.05 ± 7.45	52.34 ± 4.55	62.28 ± 0.41	64.91 ± 8.27	54.97 ± 1.65	59.94 ± 2.07	55.26 ± 2.89	73.10 ± 0.00	**75.15 ± 3.73**
6	30.62 ± 2.61	34.92 ± 1.62	43.41 ± 0.92	37.66 ± 1.56	33.61 ± 4.16	48.40 ± 0.21	48.55 ± 1.56	39.46 ± 0.71	46.65 ± 0.85	48.60 ± 0.07	45.75 ± 3.39	**55.55 ± 4.52**
7	53.39 ± 11.08	62.29 ± 8.34	75.93 ± 3.51	58.04 ± 7.02	73.54 ± 9.91	71.51 ± 0.15	65.34 ± 0.16	52.80 ± 1.10	68.18 ± 1.25	69.89 ± 0.10	73.98 ± 6.74	**82.51 ± 0.74**
8	38.46 ± 0.00	80.77 ± 0.00	90.07 ± 2.27	78.85 ± 8.16	73.08 ± 10.88	**92.31 ± 0.00**	57.69 ± 5.44	90.39 ± 2.72	90.39 ± 2.72	67.31 ± 8.16	88.47 ± 5.44	82.70 ± 2.72
9	12.04 ± 13.60	47.88 ± 0.43	45.93 ± 6.55	36.34 ± 4.72	35.33 ± 9.60	53.24 ± 0.57	38.97 ± 5.87	51.01 ± 0.86	48.99 ± 1.72	42.00 ± 2.43	**55.77 ± 2.43**	45.04 ± 2.15
10	75.82 ± 1.84	70.16 ± 1.30	74.30 ± 2.90	66.56 ± 2.13	72.80 ± 2.08	73.85 ± 0.24	73.64 ± 2.31	70.58 ± 0.59	74.39 ± 0.89	74.10 ± 2.96	74.48 ± 0.88	**76.87 ± 0.36**
11	8.79 ± 8.56	23.29 ± 1.94	28.31 ± 9.37	17.70 ± 11.79	25.57 ± 25.51	34.14 ± 0.16	11.30 ± 1.13	33.56 ± 0.97	27.74 ± 0.81	**35.16 ± 2.26**	27.74 ± 5.66	21.23 ± 15.50
12	50.00 ± 28.28	90.00 ± 0.00	78.75 ± 12.37	85.00 ± 7.07	80.00 ± 0.00	90.00 ± 0.00	80.00 ± 0.00	90.00 ± 0.00	90.00 ± 0.00	90.00 ± 0.00	95.00 ± 7.07	**100.00 ± 0.00**
13	46.74 ± 19.98	72.83 ± 4.61	64.18 ± 16.98	52.18 ± 24.60	65.22 ± 9.22	81.52 ± 1.54	56.52 ± 12.29	**83.69 ± 1.54**	71.74 ± 3.07	81.52 ± 4.61	79.35 ± 7.69	80.44 ± 12.30
OA (%)	48.06 ± 0.88	53.16 ± 3.97	58.87 ± 2.24	50.61 ± 0.49	58.01 ± 1.03	58.54 ± 0.01	55.63 ± 0.11	46.96 ± 0.56	56.59 ± 0.37	61.13 ± 0.36	58.31 ± 0.37	**63.45 ± 0.81**
AA (%)	41.97 ± 3.32	57.22 ± 0.14	58.71 ± 3.25	53.04 ± 3.42	55.61 ± 2.64	63.80 ± 0.17	53.62 ± 1.32	58.25 ± 0.21	61.85 ± 0.23	61.57 ± 0.50	63.91 ± 2.00	**64.79 ± 2.16**
Kappa×100	40.14 ± 1.25	45.68 ± 3.75	51.64 ± 2.21	43.39 ± 0.64	50.56 ± 0.21	51.96 ± 0.01	48.29 ± 0.01	39.94 ± 0.59	50.08 ± 0.37	54.68 ± 0.19	51.52 ± 0.22	**56.86 ± 0.47**

the best performance at 9×9, Augsburg City and Beijing at 11×11, and Wuhan reaches the highest accuracy at 13×13.

4) *Parameters α and β in the Dynamic Distance Threshold:* The dynamic distance threshold refines the candidate node connections selected via the cumulative probability threshold. In (17), parameter α modulates the influence of global distance statistics, controlling the threshold's responsiveness to overall distribution variations. A larger α allows the inclusion of more distant yet relevant nodes. Parameter β controls the magnitude of the nearest-neighbor adjustment term based on the normalized distance; a larger β makes the threshold more inclined to retain closer neighbors. Together, α and β enable adaptive thresholding that incorporates both global and local characteristics. As shown in Fig. 9, the optimal classification accuracy is achieved when $\alpha = 0.08$ and $\beta = 0.01$.

D. Comparison and Analysis of Results

To thoroughly evaluate the effectiveness of the proposed LKA-GFNet, we compared it with 11 SOTA algorithms: CCR-Net [44], SepDGConv [45], AM3Net [11], ExViT [46], MFT [47], MSFMamba [24], Fusion-CNN [12], S2FL [9], MDAS [48], S3F2Net [27], and DAHGMN [28]. Among them, CCR-Net, SepDGConv, AM3Net, ExViT, MFT, and MSFMamba are dual-source fusion classification methods designed for fusing HS with LiDAR or SAR data. Fusion-CNN, S2FL, MDAS, S3F2Net, and DAHGMN are tri-source fusion classification algorithms intended for the joint use of HS, MS, and LiDAR or SAR data. Notably, although S3F2Net and DAHGMN were originally designed for dual-source fusion, their fusion frameworks exhibit good extensibility and can be directly extended to tri-source fusion tasks. For a fair comparison, all competing methods were implemented with the recommended parameters reported in their original papers or official code.

Tables III–VI present the per-class classification accuracy of all algorithms across the four datasets, along with overall evaluation indicators: OA, AA, and Kappa. In terms of overall metrics, LKA-GFNet achieves the best performance on all four datasets, and its advantage is reflected not only in OA but also consistently in AA and Kappa. Taking Houston2013 as an example, LKA-GFNet attains

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

TABLE V

QUANTITATIVE COMPARISON OF DIFFERENT ALGORITHMS ON BEIJING DATASET

No.	CCR-Net	SepDGConv	AM3Net	ExViT	MFT	MSFMamba	Fusion-CNN	S2FL	MDAS	S3F2Net	DAHGMN	Proposed
1	77.25 ± 9.98	84.51 ± 0.28	**90.59 ± 3.32**	83.14 ± 6.10	85.10 ± 0.55	85.00 ± 2.36	81.47 ± 4.85	74.71 ± 1.11	80.20 ± 0.28	83.04 ± 4.58	88.93 ± 0.42	88.63 ± 0.28
2	17.91 ± 6.19	39.55 ± 6.58	22.68 ± 4.21	32.14 ± 5.99	34.49 ± 1.76	34.03 ± 3.97	39.18 ± 2.80	27.07 ± 0.39	29.14 ± 0.33	27.03 ± 8.79	**42.04 ± 3.32**	40.20 ± 1.89
3	56.11 ± 15.89	28.98 ± 2.65	58.11 ± 0.06	45.28 ± 6.72	54.25 ± 1.10	43.08 ± 2.17	43.44 ± 5.28	27.24 ± 1.36	36.10 ± 1.12	57.32 ± 5.72	56.09 ± 0.24	**67.25 ± 1.53**
4	33.61 ± 15.73	46.84 ± 3.98	**60.94 ± 3.90**	42.74 ± 5.46	38.88 ± 4.63	49.18 ± 0.33	26.81 ± 0.16	34.08 ± 0.50	44.73 ± 0.99	44.61 ± 2.82	49.18 ± 11.92	42.27 ± 5.80
5	33.62 ± 7.61	59.98 ± 0.55	**65.31 ± 14.42**	51.72 ± 6.72	55.70 ± 1.55	51.09 ± 0.99	40.33 ± 4.97	48.67 ± 0.44	52.03 ± 0.33	63.10 ± 2.76	61.47 ± 0.66	53.59 ± 1.44
6	7.47 ± 2.63	26.52 ± 3.97	35.82 ± 7.09	19.86 ± 0.79	24.49 ± 3.97	26.32 ± 0.70	18.42 ± 3.23	20.46 ± 0.45	18.84 ± 0.05	**37.02 ± 2.13**	32.85 ± 2.88	24.46 ± 4.91
7	43.17 ± 0.61	64.21 ± 4.91	71.78 ± 6.78	46.10 ± 11.50	53.36 ± 3.07	66.05 ± 1.07	50.22 ± 14.88	47.07 ± 0.62	56.51 ± 0.46	67.25 ± 3.68	67.03 ± 5.22	**72.02 ± 3.68**
8	24.20 ± 27.02	59.23 ± 2.71	**86.25 ± 8.48**	63.70 ± 3.60	59.87 ± 20.72	63.38 ± 4.05	42.04 ± 7.20	48.41 ± 0.00	64.97 ± 0.90	35.35 ± 5.86	75.80 ± 4.50	65.29 ± 4.96
9	75.00 ± 27.50	88.89 ± 0.00	73.96 ± 10.31	86.11 ± 3.93	83.34 ± 7.86	88.89 ± 0.00	77.78 ± 7.86	61.11 ± 0.00	83.33 ± 0.00	**94.44 ± 0.00**	80.56 ± 3.92	83.33 ± 0.00
10	21.95 ± 24.72	44.40 ± 1.73	65.32 ± 1.07	56.89 ± 11.34	65.44 ± 10.95	51.93 ± 2.25	51.13 ± 9.18	30.16 ± 0.30	36.71 ± 0.74	67.27 ± 9.40	54.72 ± 5.49	**71.68 ± 8.18**
11	66.67 ± 4.71	74.17 ± 1.18	72.59 ± 5.76	63.34 ± 11.79	79.17 ± 1.18	75.00 ± 0.00	50.84 ± 5.89	62.50 ± 3.54	69.17 ± 1.18	65.00 ± 7.07	80.00 ± 2.36	**80.84 ± 5.89**
12	33.33 ± 35.36	83.33 ± 0.00	**91.67 ± 11.79**	79.17 ± 5.89	66.67 ± 11.79	83.33 ± 0.00	75.00 ± 11.78	70.83 ± 5.89	66.67 ± 11.79	41.67 ± 0.00	79.17 ± 5.89	87.50 ± 5.90
13	40.00 ± 0.00	90.00 ± 14.14	63.34 ± 4.72	70.00 ± 14.14	80.00 ± 0.00	**100.00 ± 0.00**	90.00 ± 14.14	80.00 ± 0.00	**100.00 ± 0.00**	80.00 ± 0.00	**100.00 ± 0.00**	**100.00 ± 0.00**
OA (%)	37.06 ± 1.20	40.48 ± 0.88	55.72 ± 0.13	44.92 ± 2.42	50.97 ± 1.02	45.80 ± 0.24	42.16 ± 0.43	32.43 ± 0.37	38.37 ± 0.28	54.11 ± 0.26	53.80 ± 1.11	**58.41 ± 0.16**
AA (%)	40.79 ± 2.60	60.82 ± 0.63	66.03 ± 0.88	56.94 ± 2.45	60.06 ± 1.20	62.87 ± 0.93	52.82 ± 0.08	48.64 ± 0.13	56.80 ± 1.07	58.70 ± 0.39	66.75 ± 0.59	**67.47 ± 0.36**
Kappa×100	26.17 ± 0.69	32.55 ± 0.67	46.98 ± 0.39	35.69 ± 1.94	41.24 ± 0.46	37.22 ± 0.53	33.00 ± 0.34	23.91 ± 0.28	30.14 ± 0.18	44.64 ± 0.17	44.89 ± 1.15	**49.01 ± 0.24**

TABLE VI

QUANTITATIVE COMPARISON OF DIFFERENT ALGORITHMS ON WUHAN DATASET

No.	CCR-Net	SepDGConv	AM3Net	ExViT	MFT	MSFMamba	Fusion-CNN	S2FL	MDAS	S3F2Net	DAHGMN	Proposed
1	86.27 ± 1.76	84.32 ± 2.91	**88.43 ± 2.07**	87.19 ± 1.05	86.84 ± 0.45	87.62 ± 0.03	88.29 ± 1.02	82.11 ± 0.25	84.12 ± 0.08	87.56 ± 0.02	87.41 ± 0.86	88.09 ± 0.31
2	47.69 ± 8.58	32.84 ± 1.17	45.28 ± 8.85	30.61 ± 3.83	55.08 ± 7.43	33.63 ± 0.03	42.29 ± 7.43	24.69 ± 0.15	30.95 ± 3.47	**58.45 ± 3.79**	48.58 ± 4.72	55.02 ± 11.24
3	39.88 ± 14.09	42.43 ± 0.19	**60.82 ± 3.61**	47.86 ± 0.52	50.11 ± 1.91	51.27 ± 0.05	33.45 ± 4.13	29.24 ± 0.71	41.08 ± 1.48	42.00 ± 3.58	46.78 ± 0.22	56.74 ± 11.72
4	35.07 ± 1.07	39.70 ± 3.74	47.17 ± 0.53	37.46 ± 3.71	34.74 ± 0.13	42.34 ± 0.05	44.94 ± 13.03	26.61 ± 0.20	38.14 ± 1.80	42.53 ± 4.54	45.84 ± 6.08	**57.59 ± 1.77**
5	47.54 ± 0.00	90.16 ± 0.00	**98.36 ± 2.32**	90.98 ± 3.48	81.97 ± 9.27	85.25 ± 0.00	77.87 ± 1.16	71.31 ± 3.48	68.85 ± 2.32	81.15 ± 5.80	95.90 ± 1.16	92.62 ± 3.48
6	9.76 ± 7.67	31.44 ± 1.15	**53.12 ± 8.69**	31.98 ± 0.25	38.89 ± 14.37	37.34 ± 0.05	10.52 ± 0.96	26.83 ± 0.51	21.36 ± 0.83	47.74 ± 1.09	49.05 ± 1.27	51.13 ± 10.22
7	14.88 ± 16.36	87.60 ± 2.34	87.61 ± 2.34	81.41 ± 2.92	64.88 ± 5.26	87.60 ± 0.00	78.10 ± 9.93	75.21 ± 1.17	79.34 ± 4.68	83.88 ± 0.58	**91.33 ± 0.59**	89.67 ± 0.58
8	21.67 ± 8.25	85.42 ± 0.59	87.92 ± 4.12	86.67 ± 1.18	54.59 ± 13.56	86.67 ± 0.00	68.75 ± 14.73	70.42 ± 1.77	79.17 ± 3.54	81.25 ± 1.77	**89.17 ± 3.54**	82.50 ± 5.90
9	72.32 ± 6.31	87.05 ± 3.16	88.39 ± 3.79	85.71 ± 0.00	71.43 ± 0.00	83.04 ± 0.00	67.86 ± 3.79	73.66 ± 1.89	77.23 ± 0.63	87.50 ± 0.00	88.84 ± 0.64	**89.29 ± 3.78**
10	5.67 ± 1.00	76.60 ± 2.01	**87.59 ± 0.50**	76.24 ± 3.51	63.12 ± 11.03	80.14 ± 0.00	59.22 ± 7.52	57.09 ± 0.50	76.60 ± 2.01	74.82 ± 16.55	82.98 ± 1.00	86.53 ± 1.00
11	5.24 ± 6.27	37.10 ± 1.14	35.35 ± 11.98	31.18 ± 3.80	30.51 ± 18.82	39.16 ± 0.16	18.28 ± 8.74	32.80 ± 3.04	29.03 ± 1.52	35.48 ± 2.66	**48.66 ± 3.04**	46.51 ± 6.84
12	54.63 ± 15.71	**91.20 ± 0.65**	69.91 ± 11.13	88.89 ± 0.00	85.65 ± 7.20	88.89 ± 0.00	72.69 ± 0.65	79.17 ± 0.65	81.94 ± 0.65	87.50 ± 1.96	90.28 ± 0.66	90.74 ± 2.62
13	44.53 ± 12.15	93.36 ± 0.55	80.47 ± 5.53	83.59 ± 7.74	91.02 ± 2.76	84.38 ± 0.00	69.53 ± 9.94	83.59 ± 1.10	80.86 ± 0.55	84.77 ± 0.55	**93.75 ± 2.21**	90.24 ± 1.66
OA (%)	49.69 ± 0.37	52.26 ± 0.71	62.26 ± 0.19	52.75 ± 0.35	58.52 ± 0.71	55.64 ± 0.02	51.76 ± 0.32	44.11 ± 0.37	49.74 ± 0.31	60.85 ± 0.18	60.55 ± 0.17	**65.85 ± 0.35**
AA (%)	37.32 ± 1.90	67.64 ± 0.84	71.57 ± 0.37	66.14 ± 1.05	62.22 ± 0.49	68.26 ± 0.01	56.29 ± 2.98	56.36 ± 1.09	60.67 ± 0.06	68.82 ± 1.61	73.74 ± 0.62	**75.13 ± 0.13**
Kappa×100	38.97 ± 0.13	43.50 ± 0.91	54.04 ± 0.11	43.92 ± 0.46	49.49 ± 0.48	47.27 ± 0.02	42.37 ± 0.64	35.07 ± 0.47	40.91 ± 0.28	52.42 ± 0.26	52.61 ± 0.42	**58.48 ± 0.28**

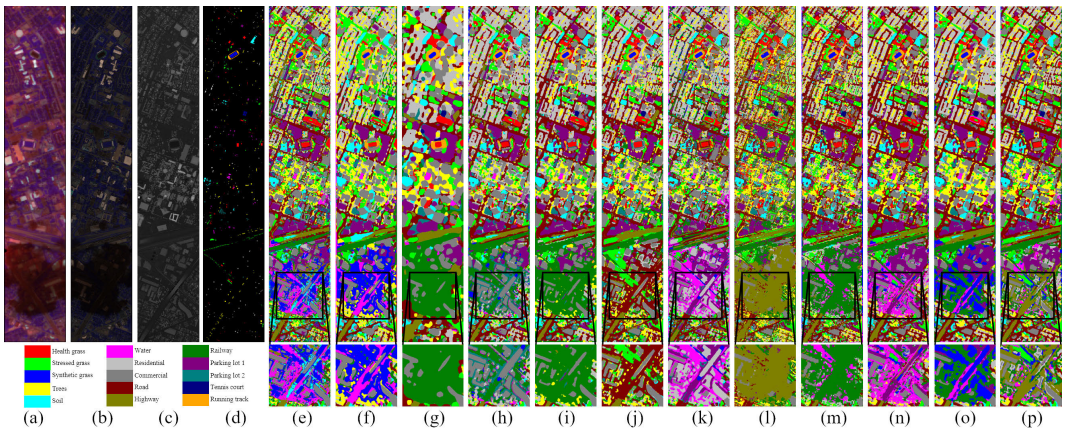


Fig. 10. Classification results on the Houston2013 dataset. (a) False-color HS image. (b) False-color MS image. (c) LiDAR image. (d) Ground truth. (e) CCR-Net (88.37%). (f) SepDGConv (83.32%). (g) AM3Net (79.88%). (h) ExViT (86.88%). (i) MFT (88.64%). (j) MSFMamba (88.99%). (k) Fusion-CNN (80.16%). (l) S2FL (84.48%). (m) MDAS (86.06%). (n) S3F2Net (91.49%). (o) DAHGMN (88.56%). (p) LKA-GFNet (92.49%).

OA/AA/Kappa of 92.49%/93.65%/91.84%, outperforming the second-best method S3F2Net by 1.00%, 1.26%, and 1.06%, respectively. In Augsburg City, LKA-GFNet achieves 63.45%/64.79%/56.86% in OA/AA/Kappa, improving upon the second-best method by 2.32%, 0.88%, and 2.18%, respectively. In Beijing, LKA-GFNet obtains OA/AA/Kappa of 58.41%/67.47%/49.01%, exceeding the second-best method by 2.69%, 0.72%, and 2.03%, respectively. In Wuhan, LKA-GFNet achieves OA/AA/Kappa of 65.85%/75.13%/58.48%, with gains of 3.59%, 1.39%, and 4.44% over the second-best method, respectively. These results demonstrate that the proposed LKA-GFNet can further reduce interclass confusion caused by cross-modal discrepancies. This advantage is also evident at the class level. LKA-GFNet yields higher accuracies on several easily confused classes. For example, in Augsburg City, it achieves 82.51% on Class 7, which is markedly higher than the second-best result of 75.93%. In the Beijing dataset, it reaches 71.68% on Class 10, still improving by 4.41% over S3F2Net (67.27%). In the Wuhan dataset, it attains 57.59% on Class 4, outperforming AM3Net (47.17%) by 10.42%. These results indicate that LKA-GFNet has stronger representation and fusion capability for fine-grained class discrimination.

We further present the classification maps on the four datasets in Figs. 10–13 for qualitative comparison in terms of spatial consistency, boundary delineation, and noise suppression. Specifically, (a)–(c) show the tri-source inputs, (d) is the ground truth, and (e)–(p) present the classification results

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

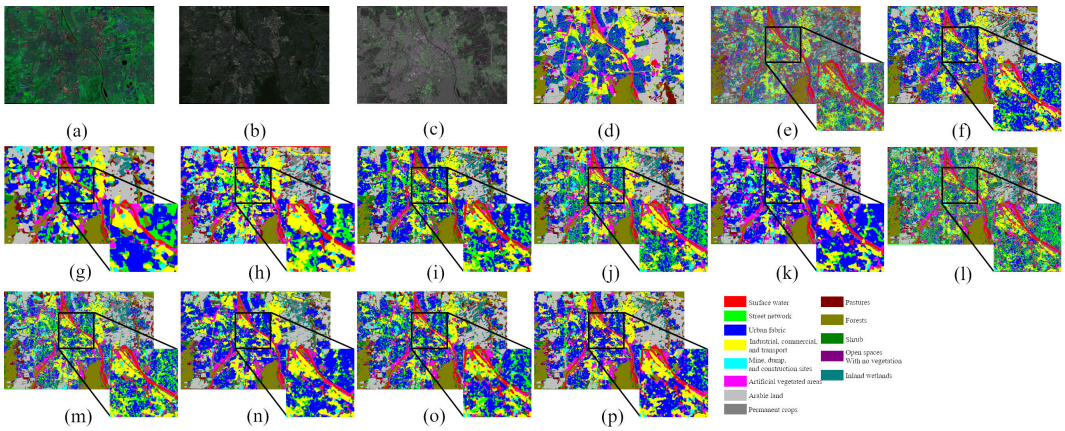


Fig. 11. Classification results on the Augsburg City dataset. (a) False-color HS image. (b) True-color MS image. (c) False-color SAR image. (d) Ground truth. (e) CCR-Net (48.06%). (f) SepDGConv (53.16%). (g) AM3Net (58.87%). (h) ExViT (50.61%). (i) MFT (58.01%). (j) MSFMamba (58.54%). (k) Fusion-CNN (55.63%). (l) S2FL (46.96%). (m) MDAS (56.59%). (n) S3F2Net (61.13%). (o) DAHGMN (58.31%). (p) LKA-GFNet (63.45%).

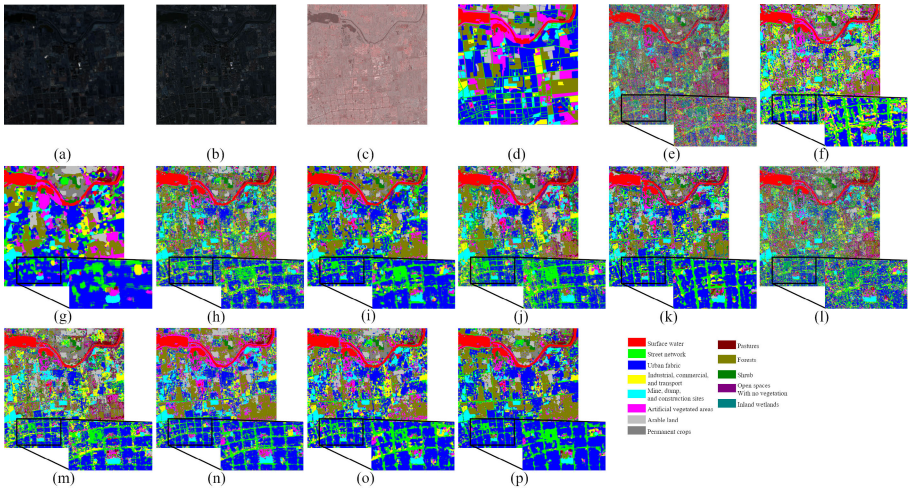


Fig. 12. Classification results on the Beijing dataset. (a) False-color HS image. (b) False-color MS image. (c) False-color SAR image. (d) Ground truth. (e) CCR-Net (37.06%). (f) SepDGConv (40.48%). (g) AM3Net (55.72%). (h) ExViT (44.92%). (i) MFT (50.97%). (j) MSFMamba (45.80%). (k) Fusion-CNN (42.16%). (l) S2FL (32.43%). (m) MDAS (38.37%). (n) S3F2Net (54.11%). (o) DAHGMN (53.80%). (p) LKA-GFNet (58.41%).

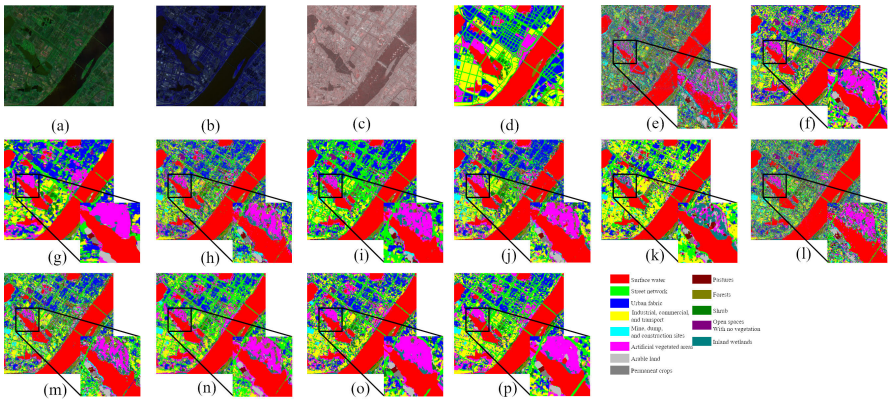


Fig. 13. Classification results on the Wuhan dataset. (a) False-color HS image. (b) False-color MS image. (c) False-color SAR image. (d) Ground truth. (e) CCR-Net (49.69%). (f) SepDGConv (52.26%). (g) AM3Net (62.26%). (h) ExViT (52.75%). (i) MFT (58.52%). (j) MSFMamba (55.64%). (k) Fusion-CNN (51.76%). (l) S2FL (44.11%). (m) MDAS (49.74%). (n) S3F2Net (60.85%). (o) DAHGMN (60.55%). (p) LKA-GFNet (65.85%).

of the competing methods and the proposed approach. In Houston2013 (see Fig. 10), the lower part of the HS false-color image contains an obvious occlusion area, which distorts the spectral information of HS data. As a result, HS+LiDAR dual-source baselines (e)–(j) tend to produce large misclassified regions and fragmented boundaries, often exhibiting

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

MODEL COMPLEXITY COMPARISON OF DIFFERENT ALGORITHMS

	CCR-Net	SepDGConv	AM3Net	ExViT	MFT	MSFMamba	Fusion-CNN	S3F2Net	DAHGMN	LKA-GFNet
Params(M)	**0.07**	2.8	2.65	0.23	0.31	0.97	0.16	0.37	0.31	0.36
FLOPs(G)	**0.00049**	2.18	36.19	2.93	1.62	12.59	0.79	0.33	3.40	1.72

TABLE VIII

COMPARISON WITH RS VISION–LANGUAGE FOUNDATION MODELS ON FOUR DATASETS

No.	Houston2013			No.	Augsburg City			Beijing			Wuhan		
	RemoteCLIP	RS5M	LKA-GFNet		RemoteCLIP	RS5M	LKA-GFNet	RemoteCLIP	RS5M	LKA-GFNet	RemoteCLIP	RS5M	LKA-GFNet
1	77.07 ± 1.41	79.96 ± 0.94	**82.48 ± 0.20**	1	49.30 ± 5.26	52.56 ± 1.97	**73.49 ± 0.00**	85.39 ± 4.85	85.78 ± 2.91	**88.63 ± 0.28**	83.14 ± 0.80	84.63 ± 1.35	**88.09 ± 0.31**
2	78.38 ± 1.20	82.33 ± 0.66	**95.44 ± 2.46**	2	**20.35 ± 4.21**	18.18 ± 6.56	8.79 ± 8.27	20.12 ± 5.79	31.58 ± 6.12	**40.20 ± 1.89**	46.57 ± 1.92	46.80 ± 17.41	**55.02 ± 11.24**
3	**99.01 ± 0.00**	**99.01 ± 0.56**	96.44 ± 0.00	3	62.13 ± 0.77	65.96 ± 6.09	**75.31 ± 12.84**	56.54 ± 1.28	51.21 ± 5.46	**67.25 ± 1.53**	52.83 ± 6.01	42.61 ± 17.64	**56.74 ± 11.72**
4	94.46 ± 0.07	**98.06 ± 0.33**	94.04 ± 3.49	4	60.24 ± 1.41	64.99 ± 8.98	**65.13 ± 8.51**	18.15 ± 7.78	28.57 ± 7.95	**42.27 ± 5.80**	41.42 ± 16.88	48.68 ± 1.54	**57.59 ± 1.77**
5	94.98 ± 1.34	98.11 ± 0.80	**100.00 ± 0.00**	5	60.23 ± 0.83	47.08 ± 6.20	**75.15 ± 3.73**	41.65 ± 15.00	49.30 ± 1.99	**53.59 ± 1.44**	71.31 ± 1.16	81.15 ± 22.02	**92.62 ± 3.48**
6	83.92 ± 0.00	86.36 ± 3.46	**100.00 ± 0.00**	6	20.18 ± 12.86	17.18 ± 4.38	**55.55 ± 4.52**	21.16 ± 3.03	14.14 ± 0.25	**24.46 ± 4.91**	14.95 ± 9.01	24.98 ± 4.41	**51.13 ± 10.22**
7	88.90 ± 0.53	**90.16 ± 0.20**	89.84 ± 0.93	7	54.08 ± 3.62	70.43 ± 4.65	**82.51 ± 0.74**	51.63 ± 5.52	61.50 ± 6.90	**72.02 ± 3.68**	80.17 ± 3.51	76.03 ± 3.51	**89.67 ± 0.58**
8	80.86 ± 0.20	78.06 ± 1.75	**93.21 ± 3.56**	8	55.77 ± 8.16	51.92 ± 2.72	**82.70 ± 2.72**	51.27 ± 13.96	36.94 ± 10.81	**65.29 ± 4.96**	61.25 ± 2.95	70.42 ± 27.70	**82.50 ± 5.90**
9	83.95 ± 1.07	83.05 ± 0.20	**87.35 ± 2.40**	9	39.68 ± 2.00	34.11 ± 10.74	**45.04 ± 2.15**	**94.44 ± 0.00**	88.89 ± 0.00	83.33 ± 0.00	86.61 ± 5.05	74.55 ± 6.94	**89.29 ± 3.78**
10	61.00 ± 0.14	61.20 ± 0.27	**82.68 ± 1.70**	10	72.46 ± 1.13	70.58 ± 4.03	**76.87 ± 0.36**	46.57 ± 1.52	63.66 ± 3.51	**71.68 ± 8.18**	63.83 ± 21.06	69.15 ± 7.52	**86.53 ± 1.00**
11	90.70 ± 0.67	92.98 ± 0.27	**96.63 ± 1.00**	11	17.24 ± 11.14	17.58 ± 3.55	**21.23 ± 15.50**	45.83 ± 12.96	50.00 ± 2.36	**80.84 ± 5.89**	16.94 ± 12.17	24.33 ± 2.09	**46.51 ± 6.84**
12	64.31 ± 0.20	73.54 ± 4.55	**95.44 ± 1.70**	12	75.00 ± 7.07	90.00 ± 14.14	**100.00 ± 0.00**	75.00 ± 0.00	58.33 ± 0.00	**87.50 ± 5.90**	78.24 ± 7.20	75.46 ± 15.06	**90.74 ± 2.62**
13	**92.98 ± 0.00**	**92.98 ± 0.00**	91.58 ± 0.49	13	70.65 ± 4.61	33.70 ± 13.83	**80.44 ± 12.30**	60.00 ± 0.00	60.00 ± 0.00	**100.00 ± 0.00**	79.30 ± 11.60	80.47 ± 5.52	**90.24 ± 1.66**
14	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	/	/	/	/	/	/	/	/	/	/
15	97.67 ± 0.30	99.05 ± 1.05	**99.58 ± 0.60**	/	/	/	/	/	/	/	/	/	/
OA (%)	83.55 ± 0.18	85.60 ± 0.27	**92.49 ± 0.22**	OA (%)	50.02 ± 0.25	54.11 ± 0.07	**63.45 ± 0.81**	44.49 ± 1.17	46.82 ± 1.71	**58.41 ± 0.16**	54.74 ± 0.06	55.64 ± 1.26	**65.85 ± 0.35**
AA (%)	85.88 ± 0.13	87.66 ± 0.01	**93.65 ± 0.18**	AA (%)	50.57 ± 1.53	48.79 ± 2.95	**64.79 ± 2.16**	51.37 ± 0.77	52.30 ± 0.55	**67.47 ± 0.36**	59.74 ± 2.68	61.49 ± 3.16	**75.13 ± 0.13**
Kappa×100	82.19 ± 0.21	84.41 ± 0.28	**91.84 ± 0.24**	Kappa×100	42.39 ± 0.30	46.24 ± 0.13	**56.86 ± 0.47**	34.09 ± 1.17	37.28 ± 1.24	**49.01 ± 0.24**	45.74 ± 0.01	46.96 ± 0.77	**58.48 ± 0.28**

severe intraregion label mixing. For example, CCR-Net and SepDGConv mainly classify this region as Water and Synthetic grass, while AM3Net and MFT assign almost the entire region to Railway. These dual-source fusion algorithms exhibit significant class interference and widespread misclassification near oblique linear structures, resulting in jagged classification boundaries. By contrast, tri-source baselines (Fusion-CNN, S2FL, MDAS, S3F2Net, and DAHGMN) use all three modalities but lack an explicit mechanism to facilitate cross-source interaction, which limits complementarity and may introduce redundancy. For example, S2FL and MDAS label a large portion of this region as Highway and Railway, which is less consistent with the typical linear morphology of these classes. Fusion-CNN and S3F2Net tend to label the region as Water and still exhibit scattered noise and broken boundaries nearby. By contrast, the proposed LKA-GFNet preserves more stable region-wise consistency and continuity in complex areas, yielding fewer misclassified regions and clearer boundaries with more natural transitions at class interfaces.

In the more complex urban scenes of Augsburg City (see Fig. 11) and Beijing (see Fig. 12), the dual-source methods (e)–(j) are more prone to boundary sticking and misclassification between built-up classes such as "urban fabric" and "industrial, commercial, and transport" and adjacent classes. This is manifested by scattered noisy patches within large regions, blurred class boundaries, or locally eroded boundaries caused by incorrect predictions. In particular, CCR-Net produces a large amount of salt-and-pepper noise. The tri-source methods (k)–(o) generally improve spatial consistency. However, some methods still leave considerable noisy points within built-up areas or suffer from class interference near road networks, resulting in less regular grid-like structures. S3F2Net and DAHGMN preserve the overall morphology relatively better. However, class sticking still persists. In contrast, LKA-GFNet achieves higher region purity and boundary continuity, effectively suppressing speckle noise while reducing adhesion between similar classes. In Wuhan (see Fig. 13), accurate delineation of water bodies, bridges, shorelines, and urban textures is particularly challenging. The dual-source baselines (e)–(j) often exhibit class spillover and broken boundaries near large water regions, where water predictions encroach on shoreline or bridge areas. Tri-source methods alleviate this issue to some extent. However, as shown in the zoomed-in views, they still fail to fully delineate both bridges. In contrast, LKA-GFNet can fully delineate the bridge structures and yields better region-wise consistency and more continuous boundaries, demonstrating clear advantages.

We also compared the computational complexity of LKA-GFNet with other DL-based approaches in terms of the number of parameters and FLOPs. As shown in Table VII, although LKA-GFNet is not the most lightweight model in terms of parameters and FLOPs, its complexity remains moderate, and its computational cost is acceptable considering the substantial improvement in classification accuracy. This indicates that LKA-GFNet achieves a favorable tradeoff between performance and complexity.

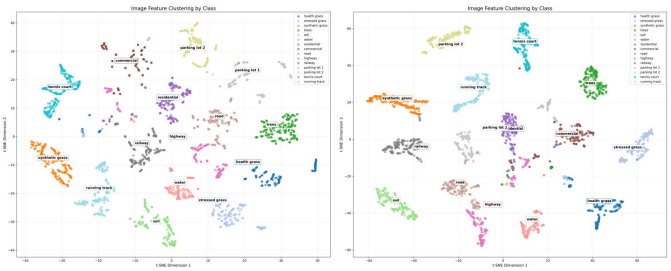
E. Comparison With RS Vision-Language Foundation Models

To further verify the effectiveness of the vision-language learning strategy adopted in the proposed LKA-GFNet, we conduct experimental comparisons with two representative modern RS vision-language foundation models, RemoteCLIP and RS5M. We follow a standard transfer-learning protocol by freezing the pretrained encoders as feature extractors and training only the task-specific classifier. Specifically, for our RS image classification task, we use the pretrained image encoders of RemoteCLIP and RS5M to extract image features and then train a linear classifier on top of these features to obtain the classification results. We follow the common protocol for CLIP-style models and therefore use only the image encoder. The text encoder is not required in this evaluation setting. Moreover, CLIP-style pretraining often yields

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
ABLATION STUDY ON THE EFFECTS OF THE PSAA AND CT-GDF MODULES

Module		Houston2013			Augsburg City			Beijing			Wuhan		
PSAA	CT-GDF	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100
×	×	89.95 ± 0.13	91.49 ± 0.23	89.11 ± 0.13	60.25 ± 0.72	62.01 ± 2.59	53.65 ± 0.70	55.70 ± 0.33	61.81 ± 2.49	44.70 ± 0.21	64.44 ± 0.06	73.19 ± 0.31	56.69 ± 0.16
✓	×	92.24 ± 0.01	93.24 ± 0.13	91.59 ± 0.02	62.20 ± 0.38	62.84 ± 2.22	55.62 ± 0.29	56.38 ± 0.40	67.12 ± 0.13	47.37 ± 0.50	65.29 ± 0.16	74.31 ± 0.27	57.51 ± 0.21
×	✓	90.81 ± 0.17	92.30 ± 0.16	90.04 ± 0.20	61.24 ± 0.89	64.46 ± 1.77	54.67 ± 0.86	56.40 ± 0.11	66.33 ± 0.41	47.24 ± 0.36	64.89 ± 0.04	74.85 ± 0.43	57.32 ± 0.18
✓	✓	**92.49 ± 0.22**	**93.65 ± 0.18**	**91.84 ± 0.24**	**63.45 ± 0.81**	**64.79 ± 2.16**	**56.86 ± 0.47**	**58.41 ± 0.16**	**67.47 ± 0.36**	**49.01 ± 0.24**	**65.85 ± 0.35**	**75.13 ± 0.13**	**58.48 ± 0.28**



(a) (b)

Fig. 14. t-SNE visualization of the high-level image feature embeddings. (a) Baseline model without text guidance. (b) Proposed LKA-GFNet with PSAA and CT-GDF.

visual features that encode useful semantic cues, making them effective for downstream visual classification. This is consistent with our setting, where LKA-GFNet performs inference using the image branch without requiring text prompts. As summarized in Table VIII, LKA-GFNet achieves the best overall performance across all four datasets compared with RemoteCLIP and RS5M.

F. t-SNE Visualization

The proposed LKA-GFNet enhances the semantic constraints of image features by introducing textual prior knowledge and graph-based fusion, which increases intraclass similarity and enhances interclass separability, thereby achieving better classification performance. To verify the improvement in feature discriminability, we perform t-SNE visualization of high-level image features of the test samples and compare the feature distributions of a baseline without text interaction and the full model with PSAA and CT-GDF. As shown in Fig. 14(a), when relying only on image features, the baseline still exhibits a certain degree of dispersion and overlap among different classes, and samples from the same class are not sufficiently compact. In contrast, after introducing PSAA and CT-GDF in Fig. 14(b), features of each class show stronger intraclass compactness in the embedding space, while the separation between different classes is further enlarged and the interclass mixing is alleviated to some extent. These results indicate that our text-guided interaction and cross-modal fusion mechanisms effectively enhance feature discriminability and promote a more structured embedding distribution.

G. Ablation Experiments

1) *Effectiveness of the PSAA and CT-GDF Modules:* In this study, we propose two core modules—PSAA and CT-GDF—for tri-source RS image fusion classification. The PSAA module utilizes geoscience textual priors to guide cross-modal feature alignment, enhancing semantic consistency and mitigating the intermodal semantic gap. This allows the model to effectively leverage high-level semantic information during feature interaction. The CT-GDF module constructs more reliable adjacency relationships through a compound threshold mechanism, facilitating adaptive integration of global statistics and local structural variations, thereby improving the robustness of graph-based decision-making. To validate the contribution of each module, we conducted ablation experiments on all four datasets. As summarized in Table IX, both modules significantly enhance classification accuracy, confirming their individual and complementary roles.
2) *SA and WA Strategies in the PSAA Module:* Within the PSAA module, we propose a dual-stage alignment strategy comprising SA and WA, aimed at balancing global semantic consistency and local fine-grained feature preservation. The SA strategy employs contrastive loss to tightly align global features of different modalities in a shared semantic space. The WA strategy uses a complementary image–text masking mechanism to build a semantic reasoning pathway that maps the tri-source image features to a common textual prototype, encouraging spontaneous alignment of tri-modal features into a unified semantic space. Ablation studies conducted across four datasets (see Table X) demonstrate that both SA and WA strategies contribute significantly to improving classification performance.
3) *Decision Fusion Strategy:* In the decision fusion strategy, we obtain three predictions, R_1, R_2, and R_3, and use majority voting to produce the final prediction. We conduct ablations to quantify the benefit of combining these outputs. Table XI presents a comparison of classification accuracy using only a single result (R_1, R_2, or R_3) versus the proposed decision fusion strategy. As shown in the table, the decision fusion strategy effectively enhances classification performance. Furthermore, within the decision fusion strategy, the refined classification result R_1 is designated as the final output in cases of disagreement among the three results. We further verify the effectiveness of this design through additional ablation, with results reported in Table XII.
4) *Text Prompt:* To verify the effectiveness of the text prompt used in LKA-GFNet, we conduct a prompt ablation study to systematically evaluate how prompt templates with different semantic complexity affect tri-source fusion classification performance. Ranging from minimal label-only prompts to prompts that gradually

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

TABLE X

ABLATION STUDY ON STRONG AND WA STRATEGIES

Alignment	Houston2013			Augsburg City			Beijing			Wuhan		
	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100
SA	91.03 ± 0.31	92.30 ± 0.47	90.29 ± 0.33	61.90 ± 0.71	64.60 ± 0.33	55.22 ± 0.63	57.02 ± 1.01	62.16 ± 1.55	46.99 ± 0.87	65.27 ± 0.06	73.94 ± 2.21	57.49 ± 0.27
WA	91.05 ± 0.80	92.54 ± 0.62	90.30 ± 0.87	62.10 ± 0.06	62.33 ± 0.06	55.38 ± 0.23	55.91 ± 1.06	65.86 ± 2.56	47.10 ± 0.32	64.63 ± 0.13	74.56 ± 0.27	56.92 ± 0.16
SA+WA	**92.49 ± 0.22**	**93.65 ± 0.18**	**91.84 ± 0.24**	**63.45 ± 0.81**	**64.79 ± 2.16**	**56.86 ± 0.47**	**58.41 ± 0.16**	**67.47 ± 0.36**	**49.01 ± 0.24**	**65.85 ± 0.35**	**75.13 ± 0.13**	**58.48 ± 0.28**

TABLE XI

ABLATION ON USING A SINGLE CLASSIFICATION RESULT VERSUS DECISION FUSION

Results	Houston2013			Augsburg City			Beijing			Wuhan		
	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100
R_1	91.33 ± 0.13	92.56 ± 0.11	90.61 ± 0.15	61.80 ± 0.31	64.61 ± 2.04	55.17 ± 0.18	57.36 ± 0.29	65.94 ± 2.56	48.15 ± 0.78	64.77 ± 0.57	73.91 ± 1.11	57.14 ± 0.35
R_2	90.81 ± 0.28	92.21 ± 0.22	90.04 ± 0.30	58.99 ± 0.51	52.46 ± 0.70	51.82 ± 0.38	56.00 ± 0.37	65.20 ± 3.28	46.71 ± 0.15	63.97 ± 0.49	72.75 ± 0.74	56.30 ± 0.45
R_3	91.19 ± 0.18	92.49 ± 0.17	90.44 ± 0.20	61.42 ± 0.16	63.49 ± 2.14	54.96 ± 0.08	55.85 ± 0.52	65.49 ± 1.94	46.50 ± 1.73	64.30 ± 0.42	72.90 ± 0.37	56.48 ± 0.33
R_1+R_2+R_3	**92.49 ± 0.22**	**93.65 ± 0.18**	**91.84 ± 0.24**	**63.45 ± 0.81**	**64.79 ± 2.16**	**56.86 ± 0.47**	**58.41 ± 0.16**	**67.47 ± 0.36**	**49.01 ± 0.24**	**65.85 ± 0.35**	**75.13 ± 0.13**	**58.48 ± 0.28**

TABLE XII

ABLATION ON THE SELECTION UNDER DISAGREEMENT

Select	Houston2013			Augsburg City			Beijing			Wuhan		
	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100
R_2	91.59 ± 0.28	93.01 ± 0.31	90.89 ± 0.30	61.06 ± 0.30	64.25 ± 0.27	54.52 ± 0.32	57.12 ± 0.29	66.46 ± 1.17	48.38 ± 0.02	65.22 ± 0.05	74.12 ± 0.84	57.66 ± 0.26
R_3	92.10 ± 0.13	93.36 ± 0.06	91.44 ± 0.14	62.15 ± 0.11	62.27 ± 0.60	55.72 ± 0.04	57.26 ± 0.11	65.32 ± 0.45	48.13 ± 0.98	65.25 ± 0.54	74.79 ± 0.28	57.69 ± 0.33
R_1	**92.49 ± 0.22**	**93.65 ± 0.18**	**91.84 ± 0.24**	**63.45 ± 0.81**	**64.79 ± 2.16**	**56.86 ± 0.47**	**58.41 ± 0.16**	**67.47 ± 0.36**	**49.01 ± 0.24**	**65.85 ± 0.35**	**75.13 ± 0.13**	**58.48 ± 0.28**

TABLE XIII

ABLATION STUDY ON THE TEXT PROMPTS

Prompts	Houston2013			Augsburg City			Beijing			Wuhan		
	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100
(1)	90.06 ± 0.39	91.36 ± 0.27	89.22 ± 0.42	61.27 ± 0.21	63.75 ± 1.02	54.60 ± 0.13	54.23 ± 0.08	66.89 ± 0.28	45.72 ± 0.11	64.63 ± 0.07	74.05 ± 0.79	57.04 ± 0.06
(2)	90.35 ± 0.33	91.92 ± 0.08	89.53 ± 0.36	61.38 ± 0.83	61.01 ± 1.23	54.35 ± 1.08	55.78 ± 0.01	65.43 ± 1.96	46.90 ± 0.30	65.35 ± 0.18	74.66 ± 0.83	57.75 ± 0.28
(3)	**92.49 ± 0.22**	**93.65 ± 0.18**	**91.84 ± 0.24**	**63.45 ± 0.81**	**64.79 ± 2.16**	**56.86 ± 0.47**	**58.41 ± 0.16**	**67.47 ± 0.36**	**49.01 ± 0.24**	**65.85 ± 0.35**	**75.13 ± 0.13**	**58.48 ± 0.28**
(4)	89.97 ± 0.01	91.61 ± 0.09	89.14 ± 0.02	61.50 ± 0.02	64.39 ± 0.62	54.96 ± 0.06	54.63 ± 0.86	67.24 ± 0.89	46.26 ± 0.52	65.80 ± 0.08	74.15 ± 0.03	58.09 ± 0.03
(5)	89.84 ± 0.11	91.53 ± 0.19	88.99 ± 0.13	61.63 ± 0.14	64.66 ± 2.04	55.17 ± 0.12	55.43 ± 0.19	65.86 ± 4.31	46.78 ± 0.59	65.09 ± 0.05	73.88 ± 1.29	57.51 ± 0.01

TABLE XIV

ABLATION STUDY ON THE DIFFERENT SOURCES

Sources	Houston2013			Augsburg City			Beijing			Wuhan		
	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100	OA (%)	AA (%)	Kappa×100
HS	83.51 ± 0.35	85.09 ± 0.03	82.16 ± 0.37	58.40 ± 0.16	57.67 ± 2.51	51.56 ± 0.30	50.79 ± 1.07	60.28 ± 0.29	40.12 ± 1.76	58.85 ± 0.45	68.68 ± 2.94	50.58 ± 0.57
MS	85.67 ± 0.02	88.28 ± 0.06	84.46 ± 0.01	57.54 ± 0.35	58.37 ± 0.18	50.25 ± 0.30	50.90 ± 0.49	63.47 ± 0.48	41.18 ± 0.11	61.28 ± 0.23	72.41 ± 0.42	53.38 ± 0.16
Radar	55.28 ± 0.51	53.24 ± 1.47	51.63 ± 0.59	39.35 ± 0.35	40.01 ± 0.01	30.62 ± 0.21	38.15 ± 0.53	41.37 ± 3.04	27.25 ± 0.71	49.00 ± 0.26	55.17 ± 1.21	38.53 ± 0.03
HS+MS	87.37 ± 0.16	89.15 ± 0.21	86.32 ± 0.17	60.96 ± 0.64	63.34 ± 2.00	54.33 ± 0.74	54.69 ± 0.01	66.57 ± 0.35	45.93 ± 0.34	63.50 ± 0.50	74.27 ± 0.83	55.67 ± 0.18
HS+Radar	88.66 ± 0.08	89.07 ± 0.55	87.71 ± 0.11	58.56 ± 0.69	60.23 ± 2.25	51.89 ± 0.89	53.13 ± 0.95	61.30 ± 1.41	43.24 ± 1.48	61.15 ± 0.47	70.59 ± 3.99	52.81 ± 0.02
MS+Radar	88.85 ± 0.10	90.88 ± 0.08	87.92 ± 0.10	59.10 ± 0.23	59.80 ± 0.40	52.19 ± 0.24	50.55 ± 0.59	62.39 ± 2.85	41.42 ± 0.21	62.43 ± 0.20	71.99 ± 0.00	54.52 ± 0.21
HS+MS+Radar	**92.49 ± 0.22**	**93.65 ± 0.18**	**91.84 ± 0.24**	**63.45 ± 0.81**	**64.79 ± 2.16**	**56.86 ± 0.47**	**58.41 ± 0.16**	**67.47 ± 0.36**	**49.01 ± 0.24**	**65.85 ± 0.35**	**75.13 ± 0.13**	**58.48 ± 0.28**

incorporate RS context and geoscientific attributes, we design five prompt templates: 1) "label."; 2) "label area."; 3) "This is label area."; 4) "This is a label area in remote sensing imagery."; and 5) "This is a label land-cover area in Earth observation imagery, linked to characteristic land-surface properties and geographic context." Here, label denotes the class name of the dataset, while the remaining text is shared across all classes to ensure a fair comparison. Table XIII shows that the prompt (3) consistently performs best on all four datasets. This suggests that adding moderate context is beneficial beyond label-only prompts, while overly descriptive prompts do not necessarily yield additional improvements. Moreover, unlike [36], which designs a dedicated prompt for each class, we adopt a fixed prompt template for all classes to ensure the generality and transferability of our method. Tri-source RS fusion datasets differ substantially in category systems, scene complexity, and spatial scales. Highly customized per-class prompt engineering may yield gains mainly from extra semantic injection or manual design, thereby obscuring the true generalization ability of the proposed method.

5) *Modality Ablation Study:* To evaluate the effectiveness of tri-source collaborative fusion and assess the contribution of each modality, we perform modality ablations with single-source, dual-source, and tri-source inputs. Single-source baselines use one modality only, dual-source baselines test all pairwise modality combinations, and the full model uses all three modalities. When a modality is ablated, we remove its corresponding branch while keeping the remaining fusion components unchanged, ensuring that performance changes are primarily due to modality availability rather than training variations. Table XIV reports the results under different modality configurations. It can be clearly observed that fusing all three sources yields the best accuracy.

V. CONCLUSION

As Earth observation enters a new era of multisensor collaboration, multisource data such as HS, MS, and radar

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

images provide more complementary information for fine-grained land-cover classification. However, existing studies still mainly focus on dual-source fusion and generally rely on image features alone, lacking a systematic way to exploit the semantic knowledge embedded in geoscience-oriented class labels. This limits the collaborative optimization of heterogeneous tri-source features under a unified semantic reference. To address these issues, we propose LKA-GFNet for tri-source fusion classification of HS, MS, and radar data. Specifically, LKA-GFNet adopts geoscience label textual priors as a shared semantic anchor and establishes a progressive strong-weak dual-stage semantic-aware alignment mechanism. This design drives structured semantic reconstruction of tri-source image feature spaces and effectively alleviates cross-modal distribution shifts. In addition, we develop a compound threshold-constrained graph decision fusion module, which adaptively constructs intramodal and cross-modal graphs using compound thresholds that combine cumulative probability and dynamic distance. By suppressing noisy connections and redundant edges while strengthening reliable information propagation, the proposed module enables more discriminative tri-source interactive fusion and robust decision-making. Extensive experiments demonstrate that LKA-GFNet achieves superior performance over existing methods on multiple public tri-source datasets in terms of overall performance, and exhibits stronger robustness to interference as well as improved boundary delineation in complex scenes. From a long-term perspective, our contribution lies not only in higher classification accuracy but also in providing a transferable learning paradigm: using unified geoscience semantic priors as a reference for cross-modal learning, we advance multisource fusion beyond traditional image-only modeling toward a dual-driven pathway that integrates both text and image. This offers a reusable framework basis for building multisource RS interpretation models with improved generalization and interpretability.

However, several issues in our method remain worthy of further investigation. For instance, the current textual prior is mainly derived from class-label semantics and has not yet fully leveraged class attributes or hierarchical relationships among classes. In addition, although the proposed graph construction strategy improves robustness, it introduces extra computational overhead in large-scale scenes, posing challenges to efficient training and inference. Therefore, our future work will focus on these issues. We will incorporate richer forms of geoscience priors, such as geoscience knowledge graphs, and combine them with learnable text prompts to enhance the fine-grained expressiveness of semantic supervision. Moreover, developing more efficient graph construction and sparse inference strategies to improve runtime efficiency, as well as investigating the performance of foundation models on our task, will also be an important direction.

ACKNOWLEDGMENT

The authors would like to express their special thanks to the organizing committee of the 2013 IEEE GRSS Data Fusion Competition for providing the hyperspectral and LiDAR data.

REFERENCES

[1] P. Dou, C. Huang, W. Han, J. Hou, Y. Zhang, and J. Gu, "Remote sensing image classification using an ensemble framework without multiple classifiers," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 190–209, Feb. 2024.

[2] J. Wang, W. Li, Y. Wang, R. Tao, and Q. Du, "Representation-enhanced status replay network for multisource remote-sensing image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 15346–15358, Nov. 2024.

[3] W. Dong, T. Yang, J. Qu, T. Zhang, S. Xiao, and Y. Li, "Joint contextual representation model-informed interpretable network with dictionary aligning for hyperspectral and LiDAR classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 11, pp. 6804–6818, Nov. 2023.

[4] X. Wang, Y. Feng, R. Song, Z. Mu, and C. Song, "Multi-attentive hierarchical dense fusion net for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 82, pp. 1–18, Jun. 2022.

[5] S. Kahraman and R. Bacher, "A comprehensive review of hyperspectral data fusion with LiDAR and SAR data," *Annu. Rev. Control*, vol. 51, pp. 236–253, Jun. 2021.

[6] V. Kumar, R. S. Singh, M. Rambabu, and Y. Dua, "Deep learning for hyperspectral image classification: A survey," *Comput. Sci. Rev.*, vol. 53, Aug. 2024, Art. no. 100658.

[7] Q. Ma, J. Jiang, X. Liu, and J. Ma, "Reciprocal transformer for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102148.

[8] B. Rasti, P. Ghamisi, J. Plaza, and A. Plaza, "Fusion of hyperspectral and LiDAR data using sparse and low-rank component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6354–6365, Nov. 2017.

[9] D. Hong, J. Hu, J. Yao, J. Chanussot, and X. X. Zhu, "Multimodal remote sensing benchmark datasets for land cover classification with a shared and specific feature learning model," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 68–80, Aug. 2021.

[10] R. Hänsch and O. Hellwich, "Fusion of multispectral LiDAR, hyperspectral, and RGB data for urban land cover classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 2, pp. 366–370, Feb. 2021.

[11] J. Wang, J. Li, Y. Shi, J. Lai, and X. Tan, "AM^{3}Net: Adaptive mutual-learning-based multimodal data fusion network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 8, pp. 5411–5426, Aug. 2022.

[12] Y. Xu, B. Du, and L. Zhang, "Multi-source remote sensing data classification via fully convolutional networks and post-classification processing," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 3852–3855.

[13] X. Pan, C. Jiao, B. Yang, H. Zhu, and J. Wu, "Attribute-guided feature fusion network with knowledge-inspired attention mechanism for multi-source remote sensing classification," *Neural Netw.*, vol. 187, Jul. 2025, Art. no. 107332.

[14] Z. Xue et al., "Multimodal self-supervised learning for remote sensing data land cover classification," *Pattern Recognit.*, vol. 157, Jan. 2025, Art. no. 110959.

[15] H. Wang, X. Liu, Z. Qiao, G. Wang, and H. Chen, "Multimodal remote sensing data classification based on Gaussian mixture variational dynamic fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5621214.

[16] J.-Y. Yang, H.-C. Li, J.-H. Yang, L. Pan, Q. Du, and A. Plaza, "Multifrequency graph convolutional network with cross-modality mutual enhancement for multisource remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505914.

[17] Z. Zhang, Y. Cai, W. Gong, X. Liu, C. Zeng, and G. Yu, "MMAGL: Multiobjective multiview attributed graph learning for joint clustering of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5505314.

[18] X. Du, X. Zheng, X. Lu, and X. Wang, "Hyperspectral and LiDAR representation with spectral–spatial graph network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 9231–9245, 2023.

[19] H. Jing et al., "Heterogeneous contrastive graph fusion network for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5521317.

[20] X. Zhao et al., "Fractional Fourier image transformer for multimodal remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 2314–2326, Feb. 2024.

[21] W.-S. Hu, W. Li, H.-C. Li, F.-H. Huang, and R. Tao, "Global clue-guided cross-memory quaternion transformer network for multisource remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 7357–7371, Apr. 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.

[22] W. Huang et al., "MCFTNet: Multimodal cross-layer fusion transformer network for hyperspectral and LiDAR data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 12803–12818, 2025.

[23] F. Zhu, C. Shi, K. Shi, and L. Wang, "Joint classification of hyperspectral and LiDAR data using hierarchical multimodal feature aggregation-based multihead axial attention transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5503817.

[24] F. Gao, X. Jin, X. Zhou, J. Dong, and Q. Du, "MSFMamba: Multiscale feature fusion state space model for multisource remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504116.

[25] M. Ma et al., "A mamba-aware spatial spectral cross-modal network for remote sensing classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4402515.

[26] T. Lu, K. Ding, W. Fu, S. Li, and A. Guo, "Coupled adversarial learning for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 93, pp. 118–131, May 2023.

[27] X. Wang, L. Song, Y. Feng, and J. Zhu, "S3F2Net: Spatial–spectral-structural feature fusion network for hyperspectral image and LiDAR data classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 5, pp. 4801–4815, May 2025.

[28] Z. Xie, L. Lv, H. Gao, S. Xu, and H. Xie, "Dual-feature attention hybrid GCN mamba network for joint hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5406514.

[29] P. Iervolino, R. Guida, D. Riccio, and R. Rea, "A novel multispectral, panchromatic and SAR data fusion for land classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 10, pp. 3966–3979, Oct. 2019.

[30] D. Hong et al., "Cross-city matters: A multimodal remote sensing benchmark dataset for cross-city semantic segmentation using high-resolution domain adaptation networks," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113856.

[31] Q. Li, F. K. K. Wong, and T. Fung, "Mapping multi-layered mangroves from multispectral, hyperspectral, and LiDAR data," *Remote Sens. Environ.*, vol. 258, Jun. 2021, Art. no. 112403.

[32] X. Meng et al., "Integrated fusion for panchromatic, multispectral, hyperspectral remote sensing images: Insights from multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5519915.

[33] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[34] F. Liu et al., "RemoteCLIP: A vision language foundation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5622216.

[35] X. Li, C. Wen, Y. Hu, and N. Zhou, "RS-CLIP: Zero shot remote sensing scene classification via contrastive vision-language supervision," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 124, Nov. 2023, Art. no. 103497.

[36] Y. Zhang, M. Zhang, W. Li, S. Wang, and R. Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501312.

[37] S. Dong, L. Wang, B. Du, and X. Meng, "ChangeCLIP: Remote sensing change detection with multimodal vision-language representation learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 53–69, Feb. 2024.

[38] X. Wang, K. Zhao, X. Zhao, and S. Li, "CSDBF: Dual-branch framework based on temporal–spatial joint graph attention with complement strategy for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540118.

[39] X. Du, X. Zheng, X. Lu, and A. A. Doudkin, "Multisource remote sensing data classification with graph fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10062–10072, Dec. 2021.

[40] T. Yang, S. Xiao, J. Qu, W. Dong, Q. Du, and Y. Li, "Graph embedding interclass relation-aware adaptive network for cross-scene classification of multisource remote sensing data," *IEEE Trans. Image Process.*, vol. 33, pp. 4459–4474, 2024.

[41] J. Chen and X. Wang, "Open set few-shot remote sensing scene classification based on a multiorder graph convolutional network and domain adaptation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4709517.

[42] T. Oorloff et al., "AVFF: Audio-visual feature fusion for video deepfake detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27102–27112.

[43] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.

[44] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, 2022.

[45] Y. Yang, D. Zhu, T. Qu, Q. Wang, F. Ren, and C. Cheng, "Single-stream CNN with learnable architecture for multisource remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5409218.

[46] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.

[47] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.

[48] J. Hu et al., "MDAS: A new multimodal benchmark dataset for remote sensing," *Earth Syst. Sci. Data*, vol. 15, no. 1, pp. 113–131, Jan. 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:58:56 UTC from IEEE Xplore. Restrictions apply.